



Antibiotic resistance in the environment: a critical insight on its occurrence, fate, and eco-toxicity

Sakina Bombaywala^{1,2} · Ashootosh Mandpe^{1,2} · Sonam Paliya^{1,2} · Sunil Kumar^{1,2}

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Abstract

The overuse, misuse, and underuse of antibiotics tend to increase the antibiotic burden in the environment resulting into the evolution in microbial community to possess resistance that renders antibiotics ineffective against them. The current review recapitulates the present state of knowledge about the occurrence and fate of antibiotics in various environmental matrices. Also, the prevalence of antibiotic-resistant bacteria/antibiotic-resistant genes (ARB/ARGs) in various biological and non-biological systems, eco-toxicity of antibiotics on non-target organisms, and remediation methods for antibiotics and ARB/ARGs removal were critically reviewed. Furthermore, a comparison of various technologies for their efficiency to eliminate antibiotic residues and ARB/ARGs is made. The study identified gaps in the investigation of toxic effects of low concentration of antibiotics and the mixture of multiple antibiotics on non-target organisms. The study of antibiotics' phytotoxicity and toxicity towards sediment and soil-dwelling organisms are also recognized as a knowledge gap. The review also details policies implemented across the globe to fight against antibiotic resistance, and the scarcity of data on lab to land transferred remediation technology was identified. The present study entails a critical review of literature providing guidelines for the articulation of policies for prudent use of antibiotics, limits on the amount of antibiotics in pharmaceutical formulations, and regular surveillance in the Indian context.

Keywords Antibiotic resistance · Horizontal gene transfer · Biodegradation · Eco-toxicity · Policy · Remediation technology

Introduction

One of the most significant examples of serendipity in science was the discovery of natural antimicrobials between

Highlights

- Globalization has led to rapid spread of antibiotic resistance genes.
- Wastewater treatment plants are inefficient in removal of ARB/ARGs.
- Scarcity of lab to land transferred remediation technologies was observed.
- Need for investigation of toxicity of antibiotics mixture on non-target organisms.
- Need to articulate policy for production, consumption, and release of antibiotics.

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✉ Sunil Kumar
s_kumar@neeri.res.in

¹ Academy of Scientific and Innovative Research (AcSIR), Ghaziabad 2010 02, India

² CSIR-National Environmental Engineering Research Institute (CSIR-NEERI), Nehru Marg, Nagpur 4400 20, India

Alexander Fleming and Ernest Duchesne. Fleming discovered penicillin in 1928; however, Ernest Duchesne was the first one to discover *Penicillium* antimicrobial properties in 1896 (Fleming 2001). A revolution in the health care sector was brought by the discovery of antibiotics that could treat microbial infections, thus saving lives worldwide; however, the community use of antibiotics far exceeds their nosocomial usage. For example, only 10% of the total antibiotic consumption in Europe was accounted for use in hospitals (Van Boeckel et al. 2014). Moreover, the load of antibiotics has increased mainly due to the unmindful use of antibiotics to treat conditions such as common cold, to which antibiotics are ineffective (CDDEP 2015; Väliälto et al. 2017). The continuous increment in the burden of antibiotics has negatively affected terrestrial and aquatic lives (Leung et al. 2012); also, it can lead to human health crises by rendering routine medical procedures life-threatening (García-Fernández et al. 2018). The groups at higher risk include people having cancer or undergoing organ-transplant and others having weakened immunity like toddlers and young children (Prestinaci et al. 2015).

The antibiotic resistance bacteria (ARB) and antibiotic resistance genes (ARGs) are acknowledged by World Health Organization (WHO) as emerging environmental pollutants and a severe threat to public health (Zarei-Baygi et al. 2019). WHO has issued a “Global Action Plan on Antimicrobial Resistance” in 2015 showing its commitment towards antibiotic resistance stewardship. The effluent from wastewater treatment plants (WWTPs), sewage, and agricultural runoffs is an additional source to the antibiotic load due to the release of partly degraded antibiotics. In the Asian region, China and India are significant contributors to the burden of antibiotics, and moreover, minimal guidelines are present in these countries for the use of antibiotics as growth promoters. These countries do not need a medical prescription for antibiotic application to be used as food for the animals (Leung et al. 2012). The effects of *globalization* on the spread of ARGs and ARBs are being recognized (MacPherson et al. 2009) due to an increase in human migration and travel, movement of food-stuff across countries, and unfettered import and export of animal commodities (Le Hello et al. 2011). Many states make food production economical by outsourcing the production of food to other developing countries, which further adds to the spread of resistance globally (Cabello et al. 2016).

Antibiotics exert their toxic activity towards the target population and also influence the non-target population with a significant impact. Freshwater algae, fishes, zooplankton, and phytoplankton are reported to be highly susceptible to the effects of antibiotics in the aquatic environment (Grenni et al. 2018).

This current article summarizes the present state of knowledge about antibiotics occurrence and fate in varied environmental habitats, including various mechanisms of antibiotic action and consequently emerging antibiotic resistance mechanisms in the microorganisms. Various studies detailing the antibiotic resistance genes and bacteria in environmental matrices in the Indian context are also reviewed to determine a possible pathway of resistance transfer. The eco-toxicity of antibiotics and antibiotic-resistant determinants on non-target organisms is also outlined. A critical review of available remediation technologies for antibiotic and antibiotic determinant elimination from WWTP effluent is also carried out in the present work.

Production and consumption of antibiotics

The easy access to medicines and an increase in prosperity have also risen the demand for antibiotics globally (Van Boeckel et al. 2015; Zhao et al. 2019). The Centers for Disease Control and Prevention (CDC) has detailed that annually about 1,000,000–2,000,000 tonnes of antibiotics are consumed (CDC 2015). A study by Klein et al. (2018) demonstrated that the consumption rate of the total global antibiotic increased by 39% between the years 2000 and 2015

(Klein et al. 2018). The mean consumption rate in 2015 for 76 countries was estimated to be 42.3 billion defined daily doses (DDDs) (Klein et al. 2018). In BRICS (Brazil, Russia, India, China, and South Africa) countries, consumption of antibiotics increased by 77% between 2000 and 2015. The utilization in 2015 was 13.5 DDDs for every 1,000 inhabitants per day (Klein et al. 2018). The country-wise increase in consumption between 2000 and 2010 in Brazil, Russia, India, China, and South Africa is 68%, 19%, 66%, 37%, and 219%, respectively (CDDEP 2015). The highest amounts of antibiotics were consumed by India in 2010, followed by China and the USA (Van Boeckel et al. 2014). The raised demand for animal protein has intensified the use of the antibiotic for prophylaxis and for growth promotion. The animal farming industry used highest quantity of antibiotics in China (23%) in 2010 followed by the USA (13%), Brazil (9%), Germany (3%), and India (3%) globally (Van Boeckel et al. 2015). In 2030, China, India, Indonesia, the USA, and Brazil are estimated to be the topmost countries to use large amounts of antibiotics in livestock (CDDEP 2015). In India, quinolones (34%) were the most frequently prescribed antibiotics in 2008, followed by cephalosporins (32%), penicillin (28%), tetracyclines (6%), and macrolides (14%) (Kotwani and Holloway 2011). China and Thailand were also noticed to follow a similar pattern of antibiotics usage (Van Boeckel et al. 2014).

Human consumption

In 2010, India, China, and the USA were reported to have the highest per capita rate of antibiotic consumption for human therapeutics (Van Boeckel et al. 2014). In the USA, the widely used antibiotic drugs in the year 2015 were tetracycline (71%), penicillin (10%), a macrolide (6%), sulfonamide (4%), aminoglycoside (4%), and lincosamide (2%). In 2015, 6.880 tonnes of tetracycline were consumed in the USA. The fluoroquinolone, cephalosporin, and amphenicol antibiotics accounted for less than 1% usage (FDA 2016).

Veterinary and aquaculture use

In animal husbandry, the antibiotics are administered to prevent and control diseases, as growth promoters, or for an increase in milk production by improving feed efficiency in dairy farms (Dibner and Richards 2005; Niewold 2007). The apiculture (beekeeping) industry also uses the antibiotics to cure disease and as growth promoters to increase production (Reybroeck et al. 2012). Globally, the highest demands are in poultry and pig raising, followed by beef cattle in the USA, Argentina, and Brazil (Kookana et al. 2014). More than 70% of antibiotics utilized in animal husbandry are for non-therapeutic purposes (Roe and Pillai 2003). Most of the antibiotics which are used for animals belong to the same antimicrobial classes and are

similar in structure and function, which are used in human medicine (Heuer et al. 2009). Tetracyclines, oxytetracycline, sulfamethazine, tylosin, penicillin G, and lincomycin are examples of antibiotics that are widely used for both animal and human medicinal purposes, such as for treatment of bacterial infections (García-Fernández et al. 2018; Kumar et al. 2019). Some antibiotics are exclusively designed for animal use only, which includes sarafloxacin and is used as prophylaxis, virginiamycin is used as a growth promoter, and monensin promotes animal growth and as milk replacer during lactation (Wegener 2003). Oxytetracycline, tylosin, and lincomycin are commonly used antibiotics in the US beekeeping sector (Reybroeck et al. 2012).

Livestock activity consumed approximately 63,200 tonnes in 2010 globally, and the highest was found in China (>15,000 tonnes). The least consuming countries were France and Canada (approximately 2 tonnes) (CDDEP 2015). The use in animal farming far outpaced human consumption (Van Boeckel et al. 2015). Approximately 15,600 tons of antibiotics, from which 80% of all the antibiotics sold in the USA between 2009 and 2015, was used for animal farming purposes (FDA 2016).

The European Union (EU) banned the avoparcin use in animal husbandry in 1997, and it also has put a restriction on the use of spiramycin, tylosin, and virginiamycin as growth promoters in 1998 (Casewell et al. 2003). Furthermore, all other growth promoter drugs used as an additive to the food supply of animals raised for its meat were banned in the EU from January 1, 2006 (European Union 2006).

The aquaculture includes the rearing of aquatic organisms like fish, aquatic plants, mollusks, and crustaceans. They are farmed in swarming and stressful environment and hence become susceptible to bacterial infections. Thus, antibiotics are administered in aquaculture for prophylactic and therapeutic use (Cabello et al. 2016). The large-scale industrialization of aquaculture has also increased the surplus demand for antibiotic production (Cabello et al. 2016). China is the largest producer (80–90%) of shrimp and carnivorous fish globally (Marshall and Levy 2011).

The extensively used antibiotics in aquaculture are oxytetracycline, florfenicol, premix, sarafloxacin, erythromycin, and sulfonamides, and the yearly consumption of florfenicol was estimated to rise from 400 kg in 2000 to 233,000 kg in 2007 (FDA 2016). The Chile salmon farming industry used fluoroquinolone ten times more than in human medicine (Asche et al. 2010).

Sources of antibiotics into the environment

Municipal sewage and hospital waste

The antibiotics are discharged from the human body via routes of urinary and/or biliary system in feces and urine in the form

of antibiotic metabolites, glucuronic and sulfuric acid conjugates, or as parent compounds (Gros et al. 2010; Milic et al. 2013; Tran et al. 2018). Approximately 70–80% of antibiotic molecules present in sewage are in consistent forms (Quoc Tuc et al. 2017). Hospital waste is also channeled into the municipal sewage system because most hospitals do not have on-site WWTP (aus der Beek et al. 2016). In European countries, 75% of antibiotics in municipal sewage are contributed by households, and only 5–20% is contributed by hospitals (Kümmerer 2009). Municipal wastewater is reported to contain from around 100–1000 ng/L of antibiotics (Li et al. 2017). Sulfamethoxazole, trimethoprim, ciprofloxacin, and ofloxacin are dominantly present in municipal wastewater (Zhang and li 2011; Aydin et al. 2019). On the contrary, β -lactams, macrolides, fluoroquinolones, sulfonamides, trimethoprim, and lincomycin are frequently detected in effluents of the hospital with the rate of detection up to 35,500 ng/L (Meena et al. 2011). The unused/expired drugs are directly disposed of in the sewage system or landfills (Rogowska et al. 2019).

A study by Barancheshme and Munir (2018) has reported WWTP influent and effluent to contain sulfamethoxazole, ofloxacin, and trimethoprim. Thus, WWTPs do not efficiently eliminate antibiotics. There is inadequate regulatory enforcement on the quality of water discharged; hence, the WWTP effluent is considered a predominant source for the dissemination of antibiotics in the environment (Aydin et al. 2019). The health issues of antibiotics present in sewage cannot be neglected due to the contamination of surface water (Ioele et al. 2016). Figure 1 depicts the sources and pathways for the flow of antibiotics in the environment.

Antibiotic production facility

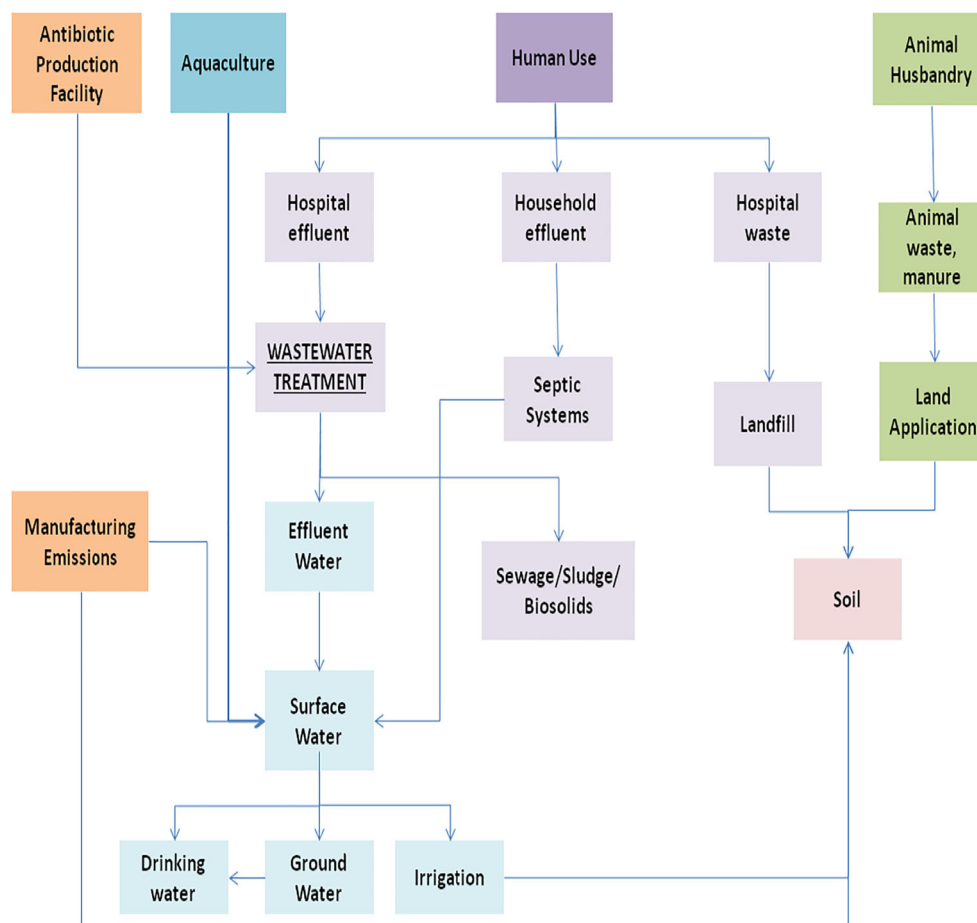
In addition to household and hospital sewage, the pharmaceutical production facilities are also a relevant contributor to antibiotic burden to the sewage treatment plant (Harrabi et al. 2018).

The effluent from the pharmaceutical industry contributes significantly to antibiotic pollution as it is not treated properly before discharge (Kookana et al. 2014). Additionally, accidental spills in the course of antibiotic manufacturing or supply are also major contamination points (Akici et al. 2018). Larsson et al. (2007) reported concentration of ciprofloxacin to be up to 3000 $\mu\text{g/L}$ in the effluent from WWTP treatment wastewater from 90 bulk pharmaceutical manufacturers in Patancheru, near Hyderabad, India.

Animal husbandry waste

Antibiotics and their metabolites have polluted the environment because of large-scale application of antibiotics in animal farming, beekeeping, and pisciculture (Hong et al. 2018).

Fig. 1 Antibiotic flow in the environment. The antibiotic in human consumption, animal food industry, aquaculture, and manufacturing are represented by purple, green, blue, and orange boxes, respectively



The route of antibiotic contamination is the unmindful release of antibiotics and dispersal of animal feces containing antibiotic metabolites as manure in agricultural fields. These antibiotics and its metabolites then reach groundwater, streams, rivers, and ditches via leaching and runoff (Mutyar and Mittal 2014). Abundantly administered virginiamycin is found in excreta of animals, and it pollutes water supply because it can bind to soil (Tasho and Cho 2016), even though its biodegradability in soil, its concentration in the soil, is enough to cause resistance in soil microbiota (Radu et al. 2017). It is noted that the degradation product of colistin sulfate in gastrointestinal region of pig had more antimicrobial activity compared to the non-degraded antibiotic (Rhouma et al. 2015). More frequently, degradation products of antibiotics are stable and long-term exposure may cause selection against resistance (Palmer et al. 2010).

Recently, it has been deduced that the vegetable crops like cabbage (Chowdhury et al. 2016), corn (Kang et al. 2013), radish (Bassil et al. 2013), and carrot (Tasho and Cho 2016) are capable of absorbing antibiotics from irrigated water. Thus, antibiotics can also flow through the food chain (Korada et al. 2018). It has been suggested that the antibiotics prevail in aquatic sediments for a sufficiently long time to cause the development of resistance in aquatic microbiota

(Fortt et al. 2007). It has been demonstrated that the antibiotics build up in fat tissues of aquatic animals at significant levels (Zhao et al. 2015). Furthermore, the resultant resistant bacteria may get transmitted to the wild population in the water environment (Marshall and Levy 2011; CDDEP 2015).

Agricultural runoff

The huge amounts of antibiotics applied to agricultural land are consequently detected in its waterways that add as a source of antibiotics to the environment (Piña et al. 2020). Additionally, the dispersal of feed and manure dust containing antibiotics is a minimal source of atmospheric contamination (Carvalho and Santos 2016).

Occurrence and fate of antibiotics

The occurrence and fate of antibiotics in the environment are determined by their structural, physical, and chemical properties (Zuccato et al. 2010; Kumar et al. 2012; Tuc Quoc Tuc et al. 2017). When exposed to temperature, pH, and light, the antibiotics become susceptible to abiotic transformation (Zhang and Li 2011; Grenni et al. 2018). Thus, these factors

govern the presence and fate of antibiotics in the environment (Bouki et al. 2013; Yan et al. 2018). The fate of most antibiotics in an aquatic ecosystem is determined by their potential to photodegradation (Yun et al. 2018). The ciprofloxacin is rapidly photodegraded at slightly basic pH, where it is present in zwitterion form. The photodegradation products at both acidic and basic pH differ, with major degradation product being a compound where the complete piperazyl ring of ciprofloxacin is replaced with an amino group at basic pH (Torniaainen et al. 1996). The half-life of sulfamethoxazole is decreased in the presence of dissolved organic matter, such as humic acid (Straub 2016). The heat, light, and extreme pH sensitize β -lactams to degradation, whereas water and methanol rapidly hydrolyze them. Therefore, β -lactams are not detected in WWTP effluent and are not found in surface waters despite their most frequent prescription around the globe (Milić et al. 2013; Tran et al. 2018). The sulfonamides are reported in WWTP effluent and seawater of Belgian harbors (Larsson 2014). The fluoroquinolone was reported in samples of soil from Hailing island and Laizhou Bay (Gao et al. 2012).

Wastewater treatment plants

WWTPs are unable to remove micro contaminants including antibiotics and thus they are released into the water bodies, sediments, and soil (Kim and Carlson 2006; He et al. 2018). The highest levels of antibiotic pollution in India were recorded from Patancheru industrial area near Hyderabad. Enormous amounts of ciprofloxacin antibiotic are detected in WWTP effluents (up to 14,000 $\mu\text{g/L}$), and lakes (2,500–6,500 $\mu\text{g/L}$) (Fick et al. 2009; aus der Beek et al. 2016). Other antibiotics such as ofloxacin and trimethoprim were also found up to 160 $\mu\text{g/L}$ and 4.4 $\mu\text{g/L}$, respectively. (Larsson et al. 2007; Fick et al. 2009).

Beta-lactams, despite their dominant presence in sewage influent, are not found significant in the effluent. It is because they are highly susceptible to acidic/enzymatic hydrolysis and, hence, are degraded during the biological treatment process (Watkinson et al. 2007). The conventional WWTP process has been reported to reduce the concentration of cephalixin, a representative of β -lactams drugs, with a removal efficiency of >96% (Costanzo et al. 2005). Administered sulfonamides are excreted as N-acetylated metabolites; their transformation back to parent compounds was reported during sewage treatment (Göbel et al. 2005). The N4-acetylsulfamethoxazole is reported to occur at 2.5–3.5 times more than sulfamethoxazole in WWTP influent (Göbel et al. 2007). The sulfonamides have low hydrophobic sorption potential ($\text{LogD}_{\text{pH}6-8}$ between -1.8 and 1.3) (Huber et al. 2005); hence, they do not get removed from sludge. This property of sulfonamides attributes to its partial removal by conventional WWTPs. After the initial lag phase of 6–12 days at 20°C following spiking, the activated sludge (AS) system was

demonstrated to biodegrade sulfonamides (Ingerslev and Halling-Sørensen 2000).

Trimethoprim is administered along with sulfonamide in the ratio 1:5, and thus, its concentration in influent is found four times lower than that of sulfamethoxazole. Likewise, in sulfonamides, the trimethoprim also does not sorb to sludge owing to its low hydrophobic partitioning coefficient (Göbel et al. 2005). The conventional WWTP processes and also AS system are not efficient in the removal of trimethoprim (Ingerslev and Halling-Sørensen 2000). However, the addition of biologically active media filtration following the secondary treatment or increasing sludge retention time (SRT) was found to be effective in trimethoprim removal (Göbel et al. 2005). Similarly, erythromycin removal improved (43 to >99%) by employing AS or aerated lagoons after secondary treatment as compared to the conventional WWTP processes (Karthikeyan and Meyer 2006).

The fluoroquinolones are removed up to 88–92% from the aqueous phase during the traditional process of WWTP (Golet et al. 2003). Also, the aminoglycoside is removed from the aqueous phase (Löffler and Ternes 2003). This can be attributed to their high hydrophobic sorption coefficient ($\text{LogD}_{\text{pH}6-8}$), and thus, they remained in digested sludge after the treatment. However, such sludge is sometimes used as a fertilizer too. Tetracycline has low $\text{LogD}_{\text{pH}6-8}$, but they tend to adsorb to biosolids through the mechanisms comprising ionic interactions, metal complexation, hydrogen bond formation, or polarization (Tolls 2001). This property of tetracycline is responsible for their removal with an efficiency of > 68% during conventional secondary treatment (Karthikeyan and Meyer 2006).

Soil

In the solid soil phase, the antibiotics undergo reversible processes of equilibrium, sorption-desorption, and sequestration. Antibiotic's polarity and water solubility control these processes. They are also a function of soil characteristics such as pH and pore size (Srinivasan et al. 2013; Figueroa-Diva et al. 2010). The bioaccessibility and bioavailability of the antibiotics are negatively correlated with their exposure time in soil. This is because micro and nano-pores in the soil are very small for the diffusion of enzymes and microorganisms (Zarfl et al. 2009; Förster et al. 2009). Antibiotics in the soil can also form non-extractable residues that result from the enzymatically catalyzed formation of covalent bonds and physical diffusion into voids (Gulkowska et al. 2013). The dissipation half-lives (DT_{50}) of antibiotics in soils are incredibly variable. Depending on soil moisture, the β -lactam antibiotics get biodegraded or abiotically transformed within hours to a few days (Braschi et al. 2013). The DT_{50} of easily extractable solids of macrolide and sulfonamide are 5–67 days and 2–10 days in soil, respectively (Carlson and Mabury

2006). The DT50 for non-extractable residues of sulfonamides reaches up to 330 days (Rosendahl et al. 2011). The half-lives for tetracyclines and fluoroquinolones extend from several months to years (Jechalke et al. 2014).

Mechanisms involved in antibiotic resistance

Misuse, underuse, and overuse of antibiotics lead to their exposure to microbes at low levels for a prolonged period (Oliveira et al. 2018). It is the rationale for the emergence and dissemination of new cases of microbial resistance, which will eventually become a component of normal bacterial metabolism. Thus, resistance genes will be active even under low concentration or in the absence (control settings) (Cruz-Loya et al. 2019). Resistance towards antibiotics can get propagated globally through horizontal gene transfer (HGT) events. The antibiotic resistance tends to raise the minimum inhibitory concentration (MIC) value of antibiotics for the microbes (Chu et al. 2018). The proliferation in antibiotic resistance is found to be positively associated with an increase in temperature and latitude coordinates. Blair et al. (2015) found that an increase of 10°C increase in temperature leads to 4.2, 2.2, and 3.2% increase in the propagation of antibiotic resistance in *Escherichia coli*, *Klebsiella pneumoniae*, and *Staphylococcus aureus*, respectively.

Prevention of access to the target

The Gram-negative bacteria are barely permeable to a lot of antibiotics since their external membrane forms a semi-permeable barrier and due to active transport by efflux pumps, as compared to Gram-positive bacteria (Pagès et al. 2008). In *Enterobacteriaceae*, the outer-membrane proteins OmpF and OmpC function as non-specific channels (Kojima and Nikaido 2013). The resistance is thus attained with downregulation of porin proteins or substitution with more-specific channels and increased efflux (Fig. 2 (1 and 2)). In *Pseudomonas* spp. and *Acinetobacter* spp., the reduction in expression of porin proteins has contributed to resistance against carbapenem and cephalosporin (Baroud et al. 2013). The resistance nodulation division (RND) pump is best described of all clinically associated Multidrug resistance (MDR) efflux transporters (Piddock 2006). They are also reported to be transmissible and are found on *IntH1* plasmid from a *Citrobacter freundii* strain (Dolejska et al. 2013). The thoroughly studied cases of RND pumps are AcrB in *E. coli* and MexB in *P. aeruginosa* (Piddock 2006). The transcription of efflux pumps is regulated by local repressors, global transcription factors, and their regulators. The mechanism of overexpression of efflux pumps includes a surge in the expression of transcription factors, MarA, SoxS, RamA, or Rob in *Enterobacteriaceae* (Bailey et al. 2010). The mutation in the

promoter (consensus −10 bp sequence upstream) of *mtrC* gene in *N. gonorrhoeae* causes genes for efflux, and multidrug resistance pumps to overexpress (Abouzeed et al. 2008) constitutively. The efflux pump genes can be induced to overexpress in response to environmental signals, while the *acrAB* genes in *E. coli* and *Salmonella* spp. are induced by indole and bile (Hirakawa et al. 2005; Baucheron et al. 2014).

Direct modification of antibiotics

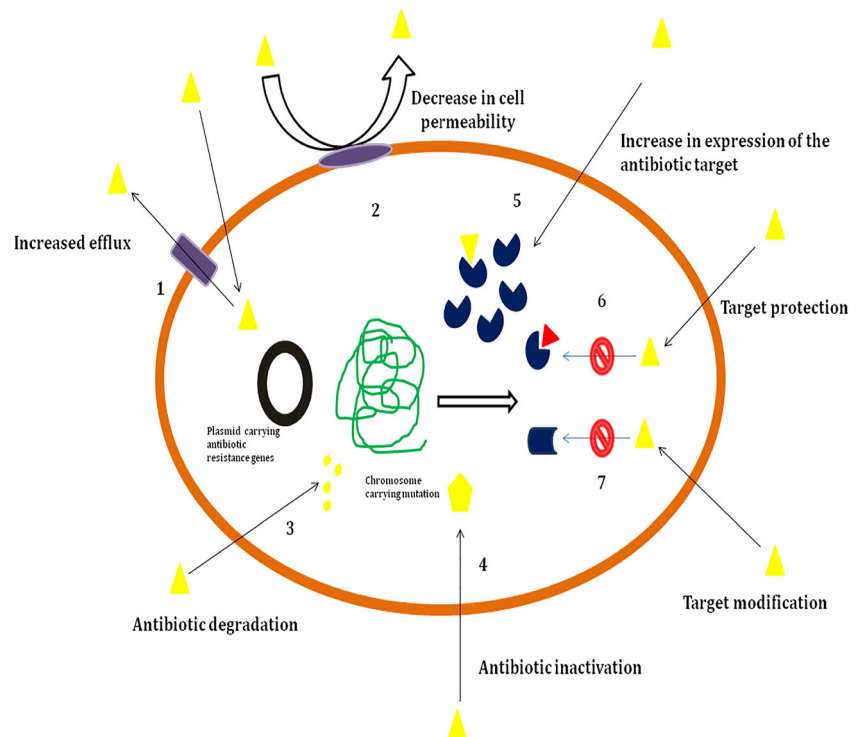
The distinct subclass of hydrolytic enzymes under the class β -lactamases confers resistance against penicillin, cephalosporins, clavams, carbapenems, and monobactams. The subclass extended-spectrum beta-lactamases (ESBLs) and carbapenemases includes imipenemase (IMP), Verona integron encoded Metallo β -lactamase (VIM), *K. pneumoniae* carbapenemase (KPC), oxacillinase (OXA), and New Delhi Metallo β -lactamase (NDM) enzymes. Hundreds of variants of CTX-M genes encode the ESBLs, and most widespread are CTX-M14 and CTX-M15 (Zhao and Hu 2013). The *kpc* and *ndm* genes encode for carbapenemase genes. The NDM carbapenemase first emerged in India in 2009 and is present in Gram-negative pathogens, such as *A. baumannii*, *K. pneumoniae*, and *E. coli* throughout the globe (Kumarasamy et al. 2010a).

Many enzymes are reported to modify aminoglycosides. These are acetyltransferases, phosphotransferases, and nucleotidyl transferases, and active sites resemble ribosomal binding cleft (Romanowska et al. 2013). A study (Qin et al. 2012) identified a novel genomic island in *Campylobacter coli* that encodes for six aminoglycoside-modifying enzymes. The rifamycin phosphotransferase confers resistance in *Actinomycetes* and pathogenic bacteria (Spanogiannopoulos et al. 2014) (Fig. 2 (3 and 4)).

Mutation in antibiotic targets

These mosaic allelic genes have arisen by recombination of DNA taken up by transformation events from the environment. A prototypical example of this is resistance to penicillin by *S. pneumoniae* is due to an enzyme that is not sensitive to penicillin and is encoded by a mosaic of penicillin-binding protein (*pbp*) gene (Dowson et al. 1993; Hakenbeck et al. 2001). The mosaicism in the *penA* gene in *N. gonorrhoeae* is associated with high degrees of resistance against extended-spectrum cephalosporins (ceftriaxone and cefixime) (Unemo et al. 2012). The *mecA* genes present in the staphylococcal cassette chromosome *mec* (SCC*mec*) element confers resistance to methicillin in methicillin-resistant *S. aureus* (MRSA). Highly divergent *mecA*, *mecI*, *mecR1*, *blaZ*, and *ccr* genes are found in SCC*mec* element that encodes for the β -lactam-insensitive protein PBP2a (Shore et al. 2011) (Fig. 2 (5)).

Fig. 2 Antibiotic resistance mechanisms. (1) Increased efflux, (2) decrease in cell permeability, (3) Antibiotic degradation, (4) antibiotic inactivation, (5) increase in expression of the antibiotic target, (6) target protection, (7) target modification



Modification (and protection) of targets

The erythromycin ribosome methylase (*erm*) gene family is known to methylate 16SrRNA, and chloramphenicol–florfenicol resistance (*cfr*) methyltransferase adds methyl to A2503 in the 23S rRNA which are reported to provide resistance against various drugs, such as macrolides, streptogramins, phenicols, lincosamides, and oxazolidinones (Long et al. 2006). The topoisomerase IV and DNA gyrase are protected from the destructive action of quinolone antibiotics by its binding with pentapeptide repeat proteins (PRPs) encoded by *qnr* genes (Vetting et al. 2011). The colistin (polymyxin B and polymyxin E) have a cyclic peptide chain with a long aliphatic side chain that causes damage to the plasma membrane of Gram-negative bacteria. The resistance to polymyxin is associated with a mutation in the two-component regulatory system PmrAB gene that results in overexpression of *pmrC* that modifies lipid A in lipopolysaccharide (LPS) (Beceiro et al. 2011). The modified LPS is not susceptible to binding by polymyxins (Fig. 2 (6 and 7)).

Prevalence of antibiotic-resistant bacteria and genes in the environment

The culture-independent techniques allow functional and structural characterization of microbial communities and enhance understanding of evolutionary pathways of ARGs spread in the environment (Allen et al. 2009). The ARGs are “easy to get, hard to lose” pollutants as they prevail in antibiotic-free

environments or when even the selective pressure has been removed (Marti et al. 2014). The antibiotic-resistant bacteria and genes prevailed in nature and have ancient origins that have no links to anthropogenic use of antibiotics (Olivares et al. 2013; Tamae et al. 2008). This “intrinsic resistome” is thus a defense mechanism of a microorganism to protect against other members of the community (Gaze et al. 2013).

The quinolone resistance genes were discovered in environmental bacteria, and only within the past 10 years there has been evidence of the mobilization of environmental resistance genes into human clinical pathogens. For example, the *qnrA* gene is present in Gram-negative bacteria *Shewanella* sp., which are found commonly in marine and freshwater environments (Poirel et al. 2005). The chromosome-encoded Qnr-like determinants present in bacteria of Vibrionaceae family are found to be homologous to *qnrB* and *qnrS* genes (Poirel et al. 2005). The antibiotics resistance genes located in human clinical pathogens are flanked by mobile genetic elements, suggesting their transfer through HGT events. Several mobile genetic elements (MGEs) involved in HGT events are presented in Table 1. The anthropogenic usage of antibiotics has likely increased the potential for gene transfer in bacterial communities and evolution in ARGs (Radu et al. 2017; Baquero et al. 2019). It is demonstrated in a study by Wright et al. (2008) that the prevalence of class 1 integrase (*intI1*) gene is reported more in the polluted river and estuarine microhabitats, and antibiotic or metal-polluted freshwater microcosms compared to that at sites with no contamination. Studies by Gillings et al. (2015) and Karkman et al. (2019)

Table 1 Summary of mobile genetic elements found in prokaryotes

Mobile genetic element	Characteristic	Example of resistance determinants	Reference
Plasmid	Variable size (1 to >100 kb), self-transmissible or mobilizable	Plasmid pP2G1 contains ARGs, which confer resistance to different antibiotics	Marti & Balca'zar, 2012
Insertion sequence	Small (<2.5Kb), contains terminal inverted repeats and encodes a transposase	IS18 mediates overexpression of the blaOXA-257 gene	Zander et al., 2013
ISCR elements	Transpose adjacent DNA sequences by rolling circle transposition	ISCR1 mediates the mobilization of the blaCTX-M-1 gene	Su et al., 2008
Transposon	Large (>12Kb) flanked by IS or inverted repeats; encodes a transposase and other functional genes such as ARGs	Tn1 and Tn3 confer resistance to b-lactams	Alekshun et al., 2007
Integron	Mediates the capture and expression of gene cassettes; encodes for an integrase, attachment sites, and transcriptional elements	Class 1 contains different cassettes conferring multidrug resistance	Marti & Balca'zar, 2012
Genomic island	Large mobile regions of DNA that encode complex biological functions	SGI1 confers resistance to streptomycin, spectinomycin, b-lactams, and sulfonamides	Levings et al., 2005
Integrating conjugative elements	MGEs that are integrated in the chromosome and transfer via conjugation	ICEVchHai1 contains different ARGs	Sjo'lund-Karlsson et al., 2011
Bacteriophages	Viruses that infect prokaryotic cells; their replication allows the DNA transfer from one bacterium to another, known as transduction	β -Lactamase genes have been isolated from environmental phages	Colomer-Lluch et al, 2011

Source: Marti et al. 2014

demonstrated the class 1 integrons to be a good proxy to the pollution of antibiotics and relationship between anthropogenic impact with abundance of *int1* gene.

In biological systems

The extended-spectrum β -lactamases (ESBL) have a nosocomial emergence. Also, methicillin-resistant *Staphylococcus aureus* (MRSA) and vancomycin-resistant *enterococci* (VRE) are frequently detected in the hospital (intensive care) setting (Otter and French 2010). Carbapenems are the last resort antibiotic against *Pseudomonas aeruginosa* and *Acinetobacter baumannii* (Brown et al. 1998). The *Klebsiella pneumoniae* carbapenemase (KPC)-type enzymes, Verona integron-encoded Metallo- β -lactamase (VIM), Imipenemase Metallo- β -lactamase (IMP), New Delhi Metallo- β -lactamase (NDM), and the OXA-48 type of enzymes are found in numerous bacterial genera throughout the world (Kumarasamy et al. 2010b; Walsh et al. 2011). The *bla*_{NDM-1} encoding for NDM has originated in the Indian subcontinent and has been disseminated to North America, the UK, and other European countries via globalization (Arya and Agarwal 2011; Walsh et al. 2011).

Human and animal gut

Several studies have shown antibiotic treatment in human-increased ARGs in the microbial flora of the human gut

(Francino 2016; Flandroy et al. 2018). The gut microbiome of regarded as a hotspot for HGT event due to high density of microbial cells and mucus that offers the environment which is favorable for conjugation (Liu et al. 2012). Additionally, the HGT is promoted in the gut by the release of membrane-destabilizing agents by microbial cells and the subsequent host inflammation (Sitaraman 2018). The experiments using mouse models reveal that the transduction in the gut is the driving force for genetic variability among gut-colonizing *E. coli* (Frazão et al. 2019). The ESBL and carbapenemases are ubiquitous in the opportunistic pathogens *E. coli* and *K. pneumoniae* and can get readily transferred in gut *Proteobacteria* (Göttig et al. 2015). The phylogenetic assignment of ARGs to microbial origin suggests that the ARGs are likely to occur in bacteria species belonging to genus *Proteobacteria* than in *Bacteroidetes*. The *tet37* gene that was originally identified from human oral metagenome together with the *ermB* gene are the two most abundant ARGs found in the human gut microbiome (Hu et al. 2013). The ARG reservoir of human commensals can be transmitted to potential human pathogen passing through the gastrointestinal tract. The food chain plays a role in the emergence of ARGs in human against antibiotics administered to animals (Bacanli and Başaran 2019). The frequency of genetic transfer is more among bacterial isolates from ecologically similar environments (Smillie et al. 2011). The study by Smillie et al. (2011) observed 25-fold more HGT between human microbiota than among ecologically diverse non-human microbiota

(Smillie et al. 2011). Still, the pathway for acquiring ARGs by gut microbiota and factors affecting ARG transfer rate is not known. Altogether, frequent HGT events in gut results in spreading of ARGs among commensals and opportunistic pathogens.

Aquatic organism

In aquaculture, the extensive usage of antimicrobial, ineffective vaccination and rearing of aqua animals at high densities leads to several deleterious changes such as a drastic reduction in the diversity of natural gut flora of fish, modification in pathogenic fish bacteria for selection of more virulent strains (Read et al. 2015), and increase of resistance levels in fish pathogens up to 60%, thereby increasing susceptibility to diseases in farmed fish and shellfish (Navarrete et al. 2008; Cabello 2006). The ARG carried by aquatic microbial community can get transmitted to pathogens of human and animals by direct contact either with ARBs residing in aquatic systems or with resistant environmental bacteria that pass ARGs to pathogens (Taylor et al. 2011). The low fitness cost of carrying ARGs in aquatic system provides potential for HGT of ARGs to pathogens (Taylor et al. 2011; Andersson and Hughes 2010). The presence of antibiotic-resistant piscine pathogens is found in *Piscirickettsia salmonis* and *Aeromonas salmonicida* from salmon aquaculture (Henríquez et al. 2016). The *tetC* gene present in the *Chlamydia suis* genome has also been transmitted from salmon pathogen *Aeromonas salmonicida* (Roberts 2009). The two macrolide-resistance genes (*mef(C)*, *mef(G)*) in *Vibrio* and *Photobacterium* might also have a marine origin (Poirel et al. 2012; Aedo et al. 2014; Nonaka et al. 2015). The dissemination of multidrug-resistant *IncA/C* plasmids from aquatic to human pathogenic *Vibrios* and *Yersinia* (Welch et al. 2009), propagation of the ARG containing *IncU* plasmid pRAS1 in *Aeromonas* pathogenic to the aquatic organism and human (Sørum et al. 2003; Cantas et al. 2013), and occurrence of plasmid-encoded quinolone resistance genes in Gram-negative bacteria (Poirel et al. 2012; Aedo et al. 2014) are examples of globalization of piscine ARGs. The secretions and stools of fish and shellfish contain antibiotic-exposed commensal and opportunistic pathogens that add to and alters microbiota of aquatic ecosystem (Navarrete et al. 2008).

Numerous microbes colonize the zooplankton and phytoplankton, and a large number of gene mutational events occur in this basal population. These shelter marine species are focal points for the addition of genetic variability and also for the emergence and spread of ARGs in the aquatic ecosystem (Jackson et al. 2011). Biofilms in the aquatic environment are present on integuments on fish and shellfish and non-living components. These biofilms are also potent hotspots for breeding and propagation of novel ARG and ARBs (Balcázar et al. 2015).

In non-biological systems

Soil: a major reservoir of the environmental resistome

It is hypothesized that the ARGs present in the soil are regarded as a store for a proportion of tomorrow's clinically relevant ARGs (Cytryn 2013). The soil sequestered antibiotic is slowly released from soil solids. The reversibility of the sequestration process makes antibiotics bio-accessible again, albeit at low concentrations (Förster et al. 2009; Zarfl et al. 2009). The antibiotics present in the soil at sub-inhibitory concentrations are an evolutionary force for the assortment of resistance genes among soil microbes (Gullberg et al. 2011). The pollution of soil by antibiotics and antibiotic resistance determinants likely increases the chance of human commensal or pathogen to acquire resistance by MGEs; for example, plasmids and integrons commute ARGs among distinctly related soil and human microbiome (Wellington et al. 2013; Klümper et al. 2015). A study by Forsberg et al. (2012) utilized high-throughput metagenomics to functionally characterize ARGs from a considerable number of isolates of multidrug-resistant *Proteobacteria* from 11 US soils. Forty-four percent of the genes from soil isolates were responsible for conferring resistance against numerous antibiotic compounds. Twenty percent of these ARG were significantly recognized as pathogenic-like multiple resistance genes. This was evinced by 90% homology of 13% of the genes and 100% homology of 7% of the genes to ARG sequences from Gram-negative and Gram-positive human pathogens. This study implied that clinically associated ARGs (*blaP1*, *tetA*, *aadB*, *sul1*, *aacA4*) are a part of multidrug resistance cassette that shuttle between soil and pathogenic bacteria and this cassette consists of ARG co-localized with long nucleotide sequences of perfect identity to bacterial host DNA, and are flanked by MGEs (Forsberg et al. 2012).

In Japan, carbapenem-resistant *Serratia marcescens* strain from hospital samples was reported to carry a novel Metallo- β -lactamase called SMB-1 on class 1 integron genetic element (Wachino et al. 2011).

The SMB-1 was found distantly linked to any medicinally associated β -lactamases; however, it had a high similarity index to a Metallo- β -lactamase termed AMO1, found in the functional metagenomic study of soil from apple orchard (Donato et al. 2010). This study is firmly evocative for soil as a reservoir for clinical ARGs and horizontal shuttle of resistance genes from an environmental strain into clinical strains. A novel class 1 integron-associated Metallo- β -lactamase gene from a clinical carbapenem-resistant *S. marcescens* strain was revealed to have high homology to allelic variant from the Alaskan soil discovered by functional metagenomic analysis (Wachino et al. 2011). This study is yet another strong evidence from the soil as a reservoir for clinical ARGs. Dantas et al. (2008) analyzed hundreds of isolates

from a large number of soil samples from farm, urban, and pristine niches, which can consume a broad range of antibiotic compounds as the sole source of carbon. Many were antibiotic-utilizing isolates and were also resistant to several antibiotics at concentrations of clinical relevance. Additionally, the bacteria belonged to diverse orders (41% *Burkholderiales*, 24% *Pseudomonadales*, 13% *Enterobacteriales*, 7% *Actinomycetales*, 7% *Rhizobiales*, and 6% *Sphingobacteriales*), and several were phylogenetically close to pathogens of human concern.

In another study, Dcosta et al. (2011) identified a vancomycin resistance gene (*vanA*) in genetic material from sediment of 30,000-year-old permafrost. These *vanA* gene variants were structurally and functionally similar to recent clinical allele. Vancomycin antibiotic is regarded as a drug of last resort, and the occurrence of this *vanA* gene variant in pristine niche validates the broad possibilities of the soil resistome to contain pathogen-associated and novel ARGs (Laskaris et al. 2010). The studies described above speculate that many clinically associated ARGs originated in antibiotic-producing soil bacteria and reached human pathogens via HGT.

Aquatic ecosystems

In the natural aquatic ecosystems, biofilms and marine sediments are relevant matrices for transfer, maintenance, and dissemination of ARGs (Huq et al. 2008). The bacterial communities in the biofilms live in a very ordered and controlled form. The high cell density and proximity in biofilms are responsible for the build-up of MGEs. Schwartz et al. (2003) suggested the transfer of the *vanA* gene to autochthonous bacteria in biofilms of wastewater and drinking water. It is demonstrated by the presence of the *vanA* gene even when *Enterococci* were not present.

Meibom et al. (2005) experimentally demonstrated an increase in transformation events in *Vibrio cholera* present in biofilm on the crab shell fragment immersed in seawater evidenced by improvement in the acquisition of resistance genes. The study thus suggested shell surface of crustaceans and mollusks as a platform for biofilm formation, and its chitin component is thought to increase bacterial competence that induces transformation uptake of extracellular DNA naturally.

The sewage impacted surface sediments along the US Mexico border, and uncontaminated sediments of a neighboring urban wetland were assessed for the occurrence of plasmid-mediated quinolone resistance (PMQR) determinants (*qnrA*, *qnrB*, *qnrS*, *qepA*, and *aac(6')-Ib-cr*). All five PMQR determinants were identified in affected sediment, whereas only three PMQR determinants were identified (*qnrB*, *qnrS*, and *qepA*) in unaffected sediment (Cummings et al. 2011). The multidrug-resistant *Pseudomonas* species isolated from aquatic sediments of Congo, India, and Switzerland overexpressed chromosomally encoded efflux pump genes.

The VIM-1 and NDM-1 are predominant resistance genes found in 29% of the *Pseudomonas* isolate from Indian sediment (Devarajan et al. 2017). A similar study showed high multiple-antibiotic resistance index and high virulence index for environmentally isolated *Pseudomonas* species suggesting it as a potential pathogen possessing high threat to humans (Singh et al. 2017).

Antibiotic resistance in rivers and lakes of India Das et al. (2020) studied the Yamuna riverine ecosystem for the presence of 139 ARGs belonging to 11 ARG types. The abundant resistance genes found conferred resistance to aminoglycoside, beta-lactam, macrolide, and tetracycline (Das et al. 2020). The ARGs in various rivers of India such as Ganga, Sabarmati, Brahmaputra, Cauvery, Kshipra, Isakavagu, and Mutha rivers have been reported. Novel ARGs have been identified in Isakavagu river, including six beta-lactam genes and one amikacin resistance genes. The most prevalent ARGs in Indian rivers are those conferring resistance to tetracycline, kanamycin, sulfonamide, levofloxacin, ciprofloxacin, and norfloxacin. A recent study has found *blaTEM* and *ampC* genes, specifically in Brahmaputra and Sabarmati river compared to its neighboring rivers in Sri Lanka (Kumar et al. 2020).

Similarly, a high abundance of *blaTEM* gene was found in the mangrove estuary of Sundarbans (Bhattacharyya et al. 2019). The discharge of untreated sewage in Yamuna river is caused by the presence of “critical pathogens” that has been listed by WHO as in urgent need of new antibiotics. Further, the high abundance of integrons, *blaTEM*, and *blaCTM* was also found in the Yamuna river (Lamba et al. 2020). Mittal et al. (2019) studied water from the Yamuna river at pre- and post-monsoon seasons. The most abundant bacterial genera in pre-monsoon were *Acinetobacter* and in post-monsoon was *Aeromonas*. The abundant ARGs belonged to efflux pump genes and Metallo-B-lactamase. The presence of *blaNDM* genes was also confirmed from both studies. The investigation of water and sediment of the Ganga river found the high prevalence of beta-lactam, multidrug efflux pumps, and elfamycin resistance genes. A study on hospital wastewater discharge in Kerala suggested possible transmission of carbapenemase-resistant *Enterobacteriaceae* in the nearby aquatic ecosystem (Kalasseril et al. 2020). The Godavari river samples also showed an increase in the count of total coliforms and fecal coliform downstream of sewage drainage sites of Nanded city, India. Similarly, a total of 60 coliforms were isolated from both upstream and downstream sites, among which 6.66% coliforms were sensitive to antibiotics used to treat human and animals infections caused by *Enterobacteriaceae* including ampicillin, nitrofurantoin, tetracycline, and ciprofloxacin and were recovered from upstream sites of Godavari river (Chitanand et al. 2010).

The functional metagenomic analysis of hypersaline and hyper alkaline Lonar lake (India) demonstrated the prevalence

of multidrug resistance efflux pumps, and beta-lactam resistance genes. Also, the presence of fluoroquinolone and acriflavine resistance genes indicates the anthropogenic contamination from sewage water of the lake (Chakraborty et al. 2020).

Waste water treatment plant influent and effluent

Sharp variations occur in the microbial composition at each step of the WWTP. The microbial fraction in wastewater adheres to organic/inorganic matter which is removed from the primary sedimentation tank. Those forming suspension or flocs of lesser density end up in the biological treatment tanks, where it is mixed with activated sludge from the secondary sedimentation tank. The arriving ARBs in raw inflow and sludge bacteria compete for organic matters in sludge. The effectiveness of the treatment on resistant removal is dictated by a shift in bacterial community dynamics during the biological treatment (Marano and Cytryn 2018). The stressors in wastewater, such as heavy metals and recalcitrant natural or human-made compounds, comprising antibiotic molecules and their metabolites, also play a role in shaping the surviving microbial community (Manaia et al. 2016; Berendonk et al. 2015). The chemical stressors, along with constant temperature and pH, large nutrient load, and a large number of bacterial cells at proximity, may accelerate HGT of MGEs conferring resistance to antibiotics, heavy metals, and other stress-causing compounds (Marano and Cytryn 2018; Di Cesare et al. 2016). The secondary effluent from biological treatment plant is estimated to contain up to 10^{12} ARB/day or 10^{18} ARGs/day (Manaia et al. 2016).

The fate of ARB/ARGs is dependent on sludge retention time (SRT) in the reactor. The high SRT will increase the proliferation of ARB/ARGs due to HGT events. The effect of the disinfection of ARB/ARGs is not entirely understood, but it is shown to reduce 2–4 log-units of the bacterial loads (Manaia et al. 2016). High doses of disinfectant are needed for inactivating ARGs, and it may also lead to severe genetic rearrangements in the microbial population (Moreira et al. 2018). The selective effect of disinfectant treatment results in a decrease in the prevalence of gene copies *per* milliliter of the sample, while it increases the prevalence of the gene copies *per* total bacteria.

In addition to stressors and nutrient loads in raw inflow, the abiotic factors in WWTPs also affect the microbial community. However, no specific abiotic factor can be used to unequivocally predict the fitness of ARB during wastewater treatment. It is expected that it might have potential to influence the development of specific group of bacteria (chemoorganoheterotrophic and facultative or aerobic bacteria), thereby conditioning the survival and proliferation of ARB. Despite these, several studies (Zhang et al. 2015; Di Cesare et al. 2016) have investigated the effect of pH and

temperature on the efficiency of sludge treatment for elimination of ARB/ARGs. The high pH and high temperature deviating from the mesophilic range are effective for removal of ARB/ARGs (Munir et al. 2011; Gao et al. 2012; Zhang et al. 2015). The association of other bio-physicochemical conditions has also been found. For instance, Di Cesare et al. (2016) observed a positive correlation between the abundance of the genes *tetA*, *ermB*, and *qnrS*, and the prokaryotic cell number, the number of bacterial aggregates, and total organic carbon. Kim et al. (2007) also concluded a relationship between total organic load, growth rate, and abundance of tetracycline-resistant bacteria in WWTP effluent. In Indian scenario, Diwan et al. (2010) reported a positive correlation between quantity of antibiotics prescribed in the hospital and level of antibiotic levels in the hospital waste water.

It is difficult to conclude a correlation between antibiotic residues found in wastewater samples and occurrence of antibiotic resistance genes. Few studies have attempted to report this cause–effect relationship (Novo et al. 2013; Nariso-Da-rocha et al. 2014; Varela et al. 2014). Narciso-Da-Rocha et al. (2014) observed a significant positive association between the prevalence of the genes *intI1*, *bla_{TEM}*, and *vanA* in the effluent and the concentration of sulfonamides and tetracycline in influent of treatment plants.

Only few recent reports have been found on efficiency of treatment plant to remove ARG/ARBs in Indian scenario. The sizeable load of multidrug resistance (MDR) bacteria was found to present in treatment effluent of hospitals of Central India (Chitnis et al. 2004). Most of the bacterial strains isolated from influent wastewater in WWTP in Pune were resistant to cefpodoxime, amoxicillin, and augmentin antibiotics (Nimonkar et al. 2019). Hospital effluent was detected with *E. coli* resistant to 12 antimicrobials (Akiba et al. 2015) in South India. Similarly, percentage of *E. coli* resistant to antibiotics increases from influent to effluent, while *E. coli* count reduced 1.06log in the effluent (Kumar et al. 2020). Three WWTPs functioning in North and Central India also reported increase in beta-lactam and tetracycline ARGs abundance in treated sewage, whereas a reduction ARB and other ARGs were observed (Saxena et al. 2021).

Eco-toxicity of antibiotic compounds

The microbial diversity operates the biogeochemical cycles in soil and water through various biological processes. Antibiotics are reported to impact these biological processes; for instance, ^{15}N tracer experiments demonstrated the impact of sulfamethazine and tetracycline on the nitrogen cycle. The increase in N_2O release and reduction in NO_3 dissimilation along with the reduction in *nirZ* and *nirS* gene potential are effects of antibiotics on the paddy field (Shan et al. 2018). Other effects of antibiotics on biogeochemical cycle include

changes in the cycling of nutrients, degradation of organic materials, sulfate reduction, and methanogenesis (Grenni et al. 2018; Xiong et al. 2019).

Often, antibiotics are reported to be toxic to non-target organisms (Kümmerer 2009). The most susceptible in the aquatic environment are primary producers and decomposers. Especially, adverse effects are caused by monensin on species diversity and biomass of phytoplankton and zooplankton (Hillis et al. 2007). The fluoroquinolones are largely present in hospital wastes and are found to be toxic to eukaryotes (Ao et al. 2018). The eco-toxicity of antibiotics is given as half-maximal effective concentration (EC_{50}). The continual exposure of the non-target organism to antibiotics has been found to result in resistance development (Haller et al. 2002).

Eco-toxicological bioassay

Standardized eco-toxicological bioassays can evaluate the acute toxicity of antibiotics on non-target organisms. The eco-toxicity is assessed by tests given in Table 2.

The test condition of eco-toxicological assays such as duration, temperature, lighting condition, and pH that might alter results of eco-toxicological bioassays ought to be considered to allow straightforward assessment of the results across both intra-species and inter-species. The mechanism of antibiotic toxicity on non-target organisms includes inhibition of chloroplast metabolisms such as synthesis of protein and photosynthesis thereby affecting cellular growth (Halling-Sorensen 2000; Nie et al. 2013). Erythromycin, ciprofloxacin, and sulfamethoxazole are reported to inhibit photosynthesis by blocking the photosystem II electron transport chain, causing the excited electrons of chloroplast to be relying on photosystem I (Nie et al. 2013). The reactive oxygen species (ROS) are formed due to that oxidative stress in a cell is

caused by the transfer of electrons from excited chlorophyll molecules in triplet state to molecular oxygen causing cellular oxidative stress. Various enzymatic and non-enzymatic antioxidants are involved in the removal of ROS and, thus, for the maintenance of redox homeostasis.

The enzymatic antioxidants are catalase, superoxide dismutase, and glutathione (GSH)-specific peroxidase, glutathione S-transferase (GST), and other enzymes of the ascorbate-GSH cycle, and non-enzymatic antioxidants are ascorbic acid (ASA) and glutathione (Nie et al. 2013). The knowledge about the generation of oxidative stress in alga in response to antibiotic exposure is limited. The study demonstrated erythromycin to be the most toxic of all the antibiotics tested on the antioxidant system of *Pseudokirchneriella subcapitata* (*P. subcapitata*). Erythromycin caused a major decline in ASA and GSH production, and reduction in enzymatic activity of ascorbate-glutathione cycle enzymes and other enzymatic antioxidants (Nie et al. 2013). In *Chlorella vulgaris*, ciprofloxacin exposure was found to affect GSH, GST, and catalase content depending on the amount of dosage (Nie et al. 2008). Similarly, on exposing *Microcystis aeruginosa* (*M. aeruginosa*) to amoxicillin, a rapid reduction in GSH concentration and activity of GST enzyme was observed, resulting due to the production of undue amounts of ROS (Liu et al. 2015).

The bacterial bioassays can be a great tool for assessing the toxicity of antibiotics in the environment. Bacteria show low acute toxicity to antibiotics compared to high toxicity of antibiotics to alga and aquatic plants. The short-term toxicity tests for an antibiotic were performed using luminescent marine bacterium *Vibrio fischeri*. The results were represented in the form of “half-maximal effective concentration (EC_{50}) values,” which is the concentration of antibiotic that led to a 50% reduction in bacterial bioluminescence. In 30-min

Table 2 Standardized eco-toxicological bioassays to evaluate the acute toxicity of antibiotics on non-target organisms

Eco-toxicological assay	Test code	Description
Respiration inhibition in activated sludge	OECD 2010	Inhibition of respiration by microorganisms in activated sludge
Toxicity effect on nitrification by microorganisms in activated sludge	ISO 9509, 1995	Assessing toxic effect on nitrification by microorganisms in activated sludge
<i>Pseudomonas putida</i> growth test	ISO 10712, 1995	The inhibition of growth is determined by turbidity measured by photometry in different dilutions of leachable soil extracts.
Luminescent inhibition of <i>Vibrio fischeri</i>	ISO 11348-2, 1998	The test measures a decrease in natural light emission of bacteria (<i>Vibrio fischeri</i>) after 30 min of exposure to contaminated soil-leachate.
<i>Daphnia Magna</i> immobilization	ISO 6341, 1996	The test determines inhibition of mobility of <i>Daphnia Magna</i> upon exposure with contaminated soil-leachate
<i>Eisenia foetida</i> mortality	(OECD 1984 DC 207, 1984)	The test assesses the number of mortality after a test duration of 14 days in contaminated soil
<i>Eisenia foetida</i> avoidance behavior	ISO/DIS 17512-1, 2008	The test evaluates the distribution behavior of earthworm (<i>Eisenia foetida</i>) in polluted soil vs control (unpolluted) soil.

Source: OECD—Organization for Economic Co-operation and Development 1984 DC 207, 1984 and OECD 2010

Microtox test, the EC₅₀ value was higher than 20 mg/L for sulfamethoxazole and oxytetracycline and more than 100 mg/L for erythromycin and ofloxacin. The EC₅₀ values were higher than 100 mg/L for ciprofloxacin and amoxicillin in 15 min of the assay (Ortiz de García et al. 2014). The toxicity towards *V. fischeri* was observed at an environmentally pertinent concentration of oxytetracycline (81 µg/L) and ofloxacin (0.014 µg/L) when assay was performed for longer duration (Ioel et al. 2016). Thus, the short-term assays such as 15-min or 30-min Microtox test and the 30-sec Microtox Flash test seem inappropriate for evaluation of the toxic effect of few antibiotics. To obtain more reliable results on antibiotic toxicity towards *V. fischeri*, the tests should be performed for a longer duration of time.

Eco-toxicity in non-target organisms

The antibiotics influence target as well as a non-target population with toxic reverberations (Grenni et al. 2018). The major shortcoming of antibiotics and other persistent organic stressors is the generation of resistance in encountered organisms (Tacconelli et al. 2018). Fluoroquinolones exert genotoxic effects on humans and are primarily used in hospital settings, where they are identified in wastewater up to 87,000 ng/l (Brown et al. 2006). The fluoroquinolones like norfloxacin and ciprofloxacin are demonstrated to be genotoxic to genetically modified *Salmonella typhimurium* at low concentration (5000 ng/l and 25,000 ng/l respectively) (Ao et al. 2018).

Chemostat study demonstrated modulation in fermentation product formation and enzyme synthesis in gut bacteria upon antibiotic addition leads to dysbiosis in the gut microbiota such *Bifidobacterium*, *Lactobacillus*, and other coliforms (such as *Clostridium*, *Escherichia coli*, *Enterococcus*). The vacant ecological niches left after antibiotic treatment might be occupied by opportunistic pathogens causing antibiotic-induced diarrhea in humans (Newton et al. 2013). The decrease in the growth of *Clostridium* upon exposure to penicillin, tylosin, ciprofloxacin and erythromycin, sulfamethoxazole, and amoxicillin is also known (Alshanti Adli 2014; Białk-Bielińska et al. 2011). The toxicity of antibiotics towards various non-target organism is presented in Table 3.

Based on the eco-toxicity assays, the “Globally Harmonized System of Classification and Labeling of Chemicals” has classified compounds as:

- i. Extremely toxic: EC₅₀ ≤ 0.1 mg/L
- ii. Very toxic: 0.1 mg/L < EC₅₀ ≤ 1 mg/L
- iii. Toxic: 1 mg/L < EC₅₀ ≤ 10 mg/L
- iv. Harmful to aquatic organisms: 10 mg/L < EC₅₀ ≤ 100 mg/L
- v. Low toxicity to aquatic organism: 100 mg/L < EC₅₀ ≤ 10000 mg/L

The toxicity of amoxicillin, erythromycin, tetracycline, levofloxacin, and norfloxacin was tested against cyanobacteria *Anabaena* CPB433 and green alga *P. subcapitata*. The cyanobacteria are prokaryotic and have a close association with pathogenic organisms (Jones et al. 2002), thus explaining its higher sensitivity to antibiotics as detected in this study. The study reported erythromycin as the most toxic antibiotic to both the organisms (green alga; EC₅₀ = 0.35 mg/l, cyanobacteria; EC₅₀ = 0.022 mg/l). The synergistic toxicity effect was highest for erythromycin and tetracycline combination in both the organisms, and risk assessment suggested ecological risk considering their actual concentrations present in the environment (González-Pleiter et al. 2013). The cyanobacteria (*M. aeruginosa*) is found to be more sensitive to both tetracycline and penicillin as (EC₅₀ = 0.09 mg/l and 0.0006 mg/l, respectively) compared to other green algae (*Selenastrum capricornutum*) (EC₅₀ = 2.2 mg/l = NOEC 100 mg/l) (Ingerslev and Halling-Sørensen 2000). Also, penicillin G was more toxic than tetracycline to cyanobacteria. Penicillin had no effect on green algae due rapid growth rate of algae that increase the pH to 9, rendering penicillin inactive. Further, the mode of action of penicillin, which is specifically the inhibition of cell wall synthesis does not affect green algae. The fluoroquinolone is 2–3 times more toxic to cyanobacterium *M. aeruginosa* than to green alga *P. subcapitata* (Robinson et al. 2005). The least described EC₅₀ values for ciprofloxacin were 0.005 mg/L (Jiang et al. 2013) and 1.1 mg/L (Yang et al. 2008) for *M. aeruginosa* and *P. subcapitata*, respectively.

Erythromycin is also found to be harmful to fish *Danio rerio* (EC₅₀ = 1000 mg/L) and to rotifers, crustaceans, and invertebrates (EC₅₀ = 20 mg/L) (Isidori et al. 2005). In another study, eco-toxicity of penicillin G, tetracycline, and vancomycin on representative organisms belonging to all trophic levels of the marine environment, that is producers (green freshwater algae *P. subcapitata*), consumers (water fleas *Daphnia magna*), and decomposers (bacteria *Vibrio fischeri*) and on edaphon (soil organisms), were assessed. The study showed green algae and *Daphnia* to be more sensitive to tetracycline than to penicillin G and vancomycin (Havelkova et al. 2016). The toxicity tests on edaphons, springtails *Folsomia candida*, and earthworms *Eisenia foetida* revealed that they were most sensitive to penicillin G. The tetracycline toxicity is not pronounced in edaphons, accounted due to its biodegradation, photodegradation, and sorption in soil (Havelkova et al. 2016). The EC₅₀ values are found to be higher (range mg/L) for invertebrates like crustaceans (*Artemia salina*, *Daphnia magna*, and *Ceriodaphnia dubia*) and cnidaria (*Hydra attenuata*) upon exposure to erythromycin, sulfamethoxazole, and oxytetracycline in a “seven-day chronic toxicity assay.” The ofloxacin and ciprofloxacin induce more acute toxicity in *Lemna minor* (duckweed) (EC₅₀ = 0.1 mg/L and 0.7 mg/L for ciprofloxacin and ofloxacin, respectively) (Robinson et al. 2005) than in *P. subcapitata* (EC₅₀ = 1.1 mg/L and 1.4 mg/L for ciprofloxacin and ofloxacin, respectively) (Isidori et al.

Table 3 Antibiotic toxicity towards non-target organisms

Test end point	Toxicity mechanism	Antibiotic class	Antibiotic	EC50 value (mg/L)	Toxicity level	Reference		
Anabaena sp. PCC 7120 strain CPB4337 with high constitutive self-luminescence								
3 day EC50	Inhibition of constitutive luminescence	Beta-lactam	Amoxicillin	56.3+ _{-2.5}	Harmful	González-Pleiter et al. 2013		
		Macrolide	Erythromycin	0.022+ _{-0.003}	Extremely toxic	González-Pleiter et al. 2013		
		Fluoroquinolone	Levofloxacin	4.8+ _{-0.4}	Toxic	González-Pleiter et al. 2013		
		Fluoroquinolone	Norfloxacin	5.6+ _{-0.5}	Toxic	González-Pleiter et al. 2013		
		Tetracycline	Tetracycline	6.2+ _{-0.8}	Toxic	González-Pleiter et al. 2013		
Pseudokirchneriella subcapitata (green algae)								
72 h EC50	Growth inhibition	Macrolide	Erythromycin	0.22	Very toxic	Isidori et al. 2005		
			Lincomycin	7.2	Toxic	Isidori et al. 2005		
			Clarithromycin	8.16	Toxic	Isidori et al. 2005		
			Clindamycin	0.014	Extremely toxic	Minguez et al. 2016		
			Azithromycin	0.5	Very toxic	Minguez et al. 2016		
			Spiramycin	4.63	toxic	Minguez et al. 2016		
			Fluoroquinolone	Ofloxacin	1.4	toxic	Yang et al. 2008	
				Norfloxacin	16.6	Harmful	Yang et al. 2008	
				Levofloxacin	7400	Low toxic	Robinson et al. 2005	
				Ciprofloxacin	1.1	Toxic	Yang et al. 2008	
				Enrofloxacin	3100	Low toxic	Robinson et al. 2005	
			Sulfonamide	Sulfamethoxazole	1.12	Toxic	Minguez et al. 2016	
				Tetracycline	Tetracycline	2.2	Toxic	Yang et al. 2008
					Oxytetracycline	0.18	Very toxic	Isidori et al. 2005
					Trimethoprim	Trimethoprim	56.01	Harmful
		Inhibition of photosynthesis		Beta-lactam	Amoxicillin	>1500	Low toxic	González-Pleiter et al. 2013
			Macrolide	Erythromycin	0.35+ _{-0.03}	Very toxic	González-Pleiter et al. 2013	
			Fluoroquinolone	Levofloxacin	>120	Harmful	González-Pleiter et al. 2013	
				Norfloxacin	>80	Harmful	González-Pleiter et al. 2013	
			Tetracycline	Tetracycline	3.31+ _{-0.96}	Toxic	González-Pleiter et al. 2013	
			Microcystis aeruginosa (freshwater cyanobacteria)					
7 day EC50	Growth inhibition	Beta-lactam	Penicillin	0.0006	Extremely toxic	Ingerslev and Halling-Sørensen 2000		
			Fluoroquinolone	Levofloxacin	7.9	Toxic	Robinson et al. 2005	
		Ciprofloxacin		0.005	Extremely toxic	Jiang et al. 2013		
		Ofloxacin		21	Harmful	Robinson et al. 2005		
		Enrofloxacin		49	Harmful	Robinson et al. 2005		
		Clinafloxacin		103	Low toxic	Robinson et al. 2005		
		Lomefloxacin		186	Low toxic	Robinson et al. 2005		
		Quinolone	Flumequine	1960	Low toxic	Robinson et al. 2005		
			Tetracycline	Tetracycline	0.09	Extremely toxic	Ingerslev and Halling-Sørensen 2000	
		Selenastrum capricornutum (green algae)						
		72 h EC50	Growth inhibition	Beta-lactam	Penicillin	2.2	Very toxic	Ingerslev and Halling-Sørensen 2000
				Tetracycline	Tetracycline	2.31	Very toxic	Ingerslev and Halling-Sørensen 2000
		Lemna minor (duckweed)						
7 day EC50	Growth inhibition	Fluoroquinolone	Levofloxacin	51	Harmful	Robinson et al. 2005		
			Ciprofloxacin	0.1	Low toxic	Robinson et al. 2005		
			Ofloxacin	126	Harmful	Robinson et al. 2005		
			Enrofloxacin	114	Harmful	Robinson et al. 2005		
			Clinafloxacin	62	Harmful	Robinson et al. 2005		
			Lomefloxacin	106	Harmful	Robinson et al. 2005		
			Quinolone	Flumequine	2470	Low toxic	Robinson et al. 2005	
			Hydra attenuate (cnidaria)					
96 h EC50	Changes in morphology	Macrolide	Erythromycin	>100	Low toxic	Kovalakova et al. 2020		
		Sulfonamide	Sulfamethoxazole	>100	Low toxic	Kovalakova et al. 2020		
		Tetracycline	Oxytetracycline	>100	Low toxic	Kovalakova et al. 2020		
Artemia salina (brine shrimp)								
72 h EC50	Mortality test	Lincosamide	Lincomycin	283	Low toxic	Migliore et al. 1997		

Table 3 (continued)

Test end point	Toxicity mechanism	Antibiotic class	Antibiotic	EC50 value (mg/L)	Toxicity level	Reference		
48 h EC50	Inhibition of hatching	Quinolone	Flumequine	308	Low toxic	Migliore et al. 1997		
		Macrolide	Clindamycin	>100	Low toxic	Minguez et al. 2016		
			Clarithromycin	>100	Low toxic	Minguez et al. 2016		
			Azithromycin	>100	Low toxic	Minguez et al. 2016		
			Spiramycin	>100	Low toxic	Minguez et al. 2016		
		Sulfonamide	Sulfamethoxazole	>100	Low toxic	Minguez et al. 2016		
		Trimethoprim	Trimethoprim	>100	Low toxic	Minguez et al. 2016		
<i>Daphnia magna</i> (Crustacean)								
48 h EC50	Inhibition of mobility	Beta-lactam	Penicillin	7114.3	Low toxic	Havelkova et al. 2016		
		Tetracycline	Tetracycline	1.82	Toxic	Havelkova et al. 2016		
		Glycopeptide	Vancomycin	370.8	Low toxic	Havelkova et al. 2016		
48 h EC50	Growth inhibition	Macrolide	Clindamycin	>100	Low toxic	Minguez et al. 2016		
			Azithromycin	>100	Low toxic	Minguez et al. 2016		
			Clarithromycin	>100	Low toxic	Minguez et al. 2016		
		Fluoroquinolone	Tylosin	46.2	Harmful	Wollenberger et al. 2000		
			Levofloxacin	>100	Low toxic	Robinson et al. 2005		
			Ciprofloxacin	>100	Low toxic	Robinson et al. 2005		
			Ofloxacin	>100	Low toxic	Robinson et al. 2005		
			Enrofloxacin	>100	Low toxic	Robinson et al. 2005		
			Clinafloxacin	>100	Low toxic	Robinson et al. 2005		
			Lomefloxacin	>100	Low toxic	Robinson et al. 2005		
			Quinolone	Flumequine	>100	Low toxic	Robinson et al. 2005	
				Oxolinic acid	4.6	Toxic	Robinson et al. 2005	
		Sulfonamide	Sulfamethoxazole	98.01	Harmful	Minguez et al. 2016		
			Sulfadiazine	13.7	Harmful	Wollenberger et al. 2000		
			Trimethoprim	Trimethoprim	>100	Low toxic	Minguez et al. 2016	
		<i>Ceriodaphnia dubia</i> (Crustacean)						
		7 day EC50	Growth inhibition	Macrolide	Erythromycin	0.94	Very toxic	Isidori et al. 2005
					Clarithromycin	12.21	Harmful	Isidori et al. 2005
				Lincosamide	Lincomycin	0.68	Very toxic	Isidori et al. 2005
Sulfonamide	Sulfamethoxazole			9.63	Toxic	Isidori et al. 2005		
Tetracycline	Oxytetracycline			1.87	Toxic	Isidori et al. 2005		
Fluoroquinolone	Ofloxacin			0.53	Very toxic	Isidori et al. 2005		
<i>Vibrio fischeri</i> (marine luminescent bacteria)								
15 min MICROTOX assay	Luminescence inhibition	Beta-lactam	Amoxicillin	>100	Low toxic	Backhaus et al. 2000		
			Clarithromycin	12.65	Harmful	Ortiz de García et al. 2014		
			Sulfonamide	Sulfamethoxazole	52.92	Harmful	Ortiz de García et al. 2014	
		Fluoroquinolone	Ciprofloxacin	>100	Low toxic	Bachkhaus et al. 2000		
			Levofloxacin	825.52	Low toxic	Ortiz de García et al. 2014		
			Ofloxacin	>100	Low toxic	Bachkhaus et al. 2000		
			Norfloxacin	20.87	Harmful	Ortiz de García et al. 2014		
		Tetracycline	Tetracycline	17.58	Harmful	Havelkova et al. 2016		
		Beta-lactam	Penicillin	>100	Low toxic	Bachkhaus et al. 2000		
			Macrolide	Clarithromycin	12.08	Harmful	Ortiz de García et al. 2014	
			Sulfonamide	Sulfamethoxazole	43.56	Harmful	Ortiz de García et al. 2014	
		Fluoroquinolone	Ofloxacin	0.000014	Extremely toxic	Ioele et al. 2016		
			Levofloxacin	788.3	Low toxic	Ortiz de García et al. 2014		
			Norfloxacin	0.022	Extremely toxic	Bachkhaus et al. 2000		
			Tetracycline	Tetracycline	6.7	Toxic	Havelkova et al. 2016	
<i>Folsomia candida</i> (springtail)								
28 days EC50	Inhibition of reproduction	Beta-lactam	Penicillin	328	low toxic	Havelkova et al. 2016		
		Tetracycline	Tetracycline	2560	Low toxic	Havelkova et al. 2016		
		Glycopeptide	Vancomycin	546	Low toxic	Havelkova et al. 2016		
<i>Eisenia fetida</i> (redworm)								
56 days EC50	Inhibition of reproduction	Beta-lactam	Penicillin	348	Low toxic	Havelkova et al. 2016		
		Tetracycline	Tetracycline	2735	Low toxic	Havelkova et al. 2016		
<i>Brachionus calyciflorus</i> (rotifer)								
48 h EC50	Growth inhibition	Macrolide	Erythromycin	0.02	Extremely toxic	Isidori et al. 2005		
		Macrolide	Clarithromycin	0.002	Extremely toxic			

Table 3 (continued)

Test end point	Toxicity mechanism	Antibiotic class	Antibiotic	EC50 value (mg/L)	Toxicity level	Reference
		Sulfonamide	Sulfamethoxazole	0.52	Very toxic	Isidori et al. 2005
		Lincosamide	Lincomycin	0.07	Extremely toxic	Isidori et al. 2005
		Fluoroquinolone	Ofloxacin	1.44	Very toxic	Isidori et al. 2005
		Tetracycline	Oxytetracycline	0.17	Very toxic	Isidori et al. 2005
<i>Anas platyrhynchos</i> (mallard duck)						
8 days EC50	Mortality test	Tetracycline	Oxytetracycline	>5620	Low toxic	Kumar et al. 2005
5 days EC50	Mortality test	Polyether Antibiotic	Monensin	>5000	Low toxic	Kumar et al. 2005
<i>Colinus virginianus</i> (northern bobwhite)						
8 days EC50	Mortality test	Tetracycline	Oxytetracycline	>5620	Low toxic	Kumar et al. 2005
<i>Lumbricina</i> (earthworm)						
5 days EC50	Mortality test	Polyether Antibiotic	Monensin	>10	Harmful	Kumar et al. 2005
		Fluoroquinolone	Sarafloxacin	1000	Low toxic	Kumar et al. 2005
28 days EC50	Mortality test	Macrolide	Tylosin	918	Low toxic	Kumar et al. 2005
<i>Skeletonema marinoi</i> (marine algae)						
72 h EC50	Growth inhibition	Macrolide	Clindamycin	154 × 10 ⁻⁶	Extremely toxic	Minguez et al. 2016
			Clarithromycin	152 × 10 ⁻⁶	Extremely toxic	Minguez et al. 2016
			Azithromycin	0.214	Very toxic	Minguez et al. 2016
			Spiramycin	0.439	Very toxic	Minguez et al. 2016
		Sulfonamide	Sulfamethoxazole	5.35	Toxic	Minguez et al. 2016
		Trimethoprim	Trimethoprim	3.19	Toxic	Minguez et al. 2016
WWTP activated sludge						
	Respirometry test	Beta-lactam	Amoxicillin	2423.29	Low toxic	Ortiz de García et al. 2014
		Macrolide	Erythromycin	561.91	Low toxic	Ortiz de García et al. 2014
			Clarithromycin	36.4	Harmful	Ortiz de García et al. 2014
		Fluoroquinolone	Ciprofloxacin	9.55	Toxic	Ortiz de García et al. 2014
			Levofloxacin	2021.41	Low toxic	Ortiz de García et al. 2014
			Norfloxacin	43.81	Harmful	Ortiz de García et al. 2014
		Sulfonamide	Sulfamethoxazole	279.81	Low toxic	Ortiz de García et al. 2014
<i>Danio rerio</i> (fish)						
96 h EC50	Inhibition of reproduction	Macrolide	Erythromycin	1000	Low toxic	Isidori et al. 2005
<i>Brachydanio rerio</i> (zebrafish)						
96 h EC50	Inhibition of reproduction	Fluoroquinolone	Levofloxacin	>100	Low toxic	Robinson et al. 2005
			Ciprofloxacin	>100	Low toxic	Robinson et al. 2005
			Ofloxacin	>100	Low toxic	Robinson et al. 2005
			Enrofloxacin	>100	Low toxic	Robinson et al. 2005
			Clinafloxacin	>100	Low toxic	Robinson et al. 2005
			Lomefloxacin	>100	Low toxic	Robinson et al. 2005
		Quinolone	Flumequine	>100	Low toxic	Robinson et al. 2005
<i>Pimephales promelas</i> (fathead minnow)						
96 h EC50	Inhibition of reproduction	Fluoroquinolone	Clinafloxacin	>10	Harmful	Robinson et al. 2005

2005; Yang et al. 2008). The ciprofloxacin and ofloxacin are highly toxic to the organisms of the water environment. Their toxicity is ranked as very toxic towards photosynthetic organisms and less toxicity towards aquatic plants and algae. Severe antibiotic toxicity to fish is only reported at a high concentration of antibiotics (Robinson et al. 2005; Minguez et al. 2016). The European Medicines Evaluation Agency (EMA) unambiguously prescribes the usage of cyanobacteria as a sensitive detection tool for testing toxicity to antibiotics in the environment (EMA 2006). Since the doubling time (growth rate) of cyanobacteria is more, the toxicity screening test duration is required to prolong up to 7 days. Additionally, the more

extended exposure period may lower EC₅₀ values for antibiotics, resulting in a decrease in sensitivity of the test (Robinson et al. 2005). To overcome this drawback, the inhibition of photosynthesis can be evaluated rather than retardation of growth in cyanobacteria by reducing the test duration to 24 h (van der Grinten et al. 2010). Cyanobacteria are found to be vulnerable to exposure to even micrograms/liter of concentrations of fluoroquinolones and amoxicillin, which is comparable to their concentrations found in the environment (Ingerslev and Halling-Sørensen 2000; Robinson et al. 2005; aus der Beek et al. 2016).

Remediation technologies for removal of antimicrobial residues, resistance bacteria, and genes

The novel treatment strategies are required to combat the rapid spreading of antibiotic resistance among pathogens. The remediation strategies that have shown the potential of their removal are adsorption mechanisms, bioelectrochemical methods, hydrolysis, redox reactions, acidic, alkaline or ionic, metal-assisted, photolysis, phytolysis, microbial, and enzymatic degradation (Massé et al. 2014; Yan et al. 2018; Dangi et al. 2019). The advanced technologies such as chlorination, ozonation (Ikehata et al. 2006; Sharma 2008), activated carbon filtration (Cong et al. 2013), membrane processes, advanced oxidation processes (AOPs) (Ikehata et al. 2006), use of nanomaterials (Aruguete et al. 2013), and ferrate treatment (Sharma 2008) have also been introduced.

Biological treatment

Application of microbes or enzymes for degradation or transformation of antibiotics might be a probable biological method to lessen the load of chemicals in the water or soil environment. An example of enzymatic degradation is the use of cytochrome P450 enzymes for antibiotic inactivation via transformation (Bhattacharya and Yadav 2017). Both biotic and abiotic processes have been shown to degrade antibiotics. Biotic degradation utilizes microorganisms, whereas degradation involves the mechanism of sorption, hydrolysis, photolysis, oxidation, and reduction reactions (Massé et al. 2014). Microbes that aid in elimination and amendment are investigated for treating pollutants in terrestrial and aquatic ecosystems. The bioremediation process is the metabolic capacity exhibited by microorganisms (Dangi et al. 2019).

Recent studies confirmed the parallelism between the antibiotic degradation capability and the microorganisms studied for antibiotic bioremediation which possesses antibiotic resistance genes. Conversely, Wen et al. (2018) reported no connection in the doxycycline degradation and resistance genes in *Candida* sp. and *Escherichia* sp. Hence, these microbes can be explored as an eco-friendly alternative for genetic engineering of antibiotic degrading genes. The bioelectrochemical system (BES) is mostly operated in microbial–fuel cell mode, which employs biological anode and abiotic cathode. The abiotic cathode consists of electron acceptors, namely, potassium ferricyanide or oxygen. The antibiotic degrading bacteria and exoelectrogenic microbes get attached to bio-anode in biofilms, and antibiotics act as a source of electrons and carbon. The *Proteobacteria*, *Firmicutes*, and *Bacteroidetes* are dominating phylum of bacteria in BES. The antibiotics added to BES changes bacteria present at the genus level.

In BES, the functional microorganisms capable of degrading sulfamethazalone, oxytetracycline chloramphenicol, and nitrofurazone are reported to be *Methanobacterium*, *Eubacterium*, *Pseudomonas*, and *Klebsiella*, respectively. The essential mineralization pathways in the BES are acetylation, nitro reduction, hydrolysis, and meta-cleavage pathways. Various factors such as pH, temperature, salinity, applied potential, and antibiotic concentration are capable of altering microbial composition and reactions in BES. Hence, an optimized BES can be designed by employing the understanding of ARBs, ARGs, microbes, and redox reactions for combating antibiotic contamination (Yan et al. 2018). WWTPs do not efficiently remove antibiotics by adsorption on activated sludge and biodegradation that leads to the release of antibiotic to receiving environments (Jones et al. 2002). Erythromycin remediation in wastewater treatment is most difficult, as suggested by its removal rate of less than 5% (Zuccato et al. 2010). The tetracyclines, sulfonamides, and removal rates of fluoroquinolones varied from 30 to 80% (Gros et al. 2010; Zuccato et al. 2010).

Du et al. (2015) conducted a study using WWTP that consisted of the anaerobic-anoxic-aerobic membrane-based reactor (MBR) process. In the anaerobic and anoxic conditions, the microorganisms have lower bioactivity, which accounts for lesser propagation of resistance genes. Thus, the study concluded that ARGs could be successfully removed in anaerobic and anoxic conditions as compared to aerobic conditions. Also, MBR is efficient as the retention of microorganisms gets affected by filtration through membranes (Du et al. 2015). Metagenomic studies have shown the anaerobic-aerobic sequence (AAS) bioreactors to be efficient and lesser energy utilizing treatment alternatives. The anaerobic treatment is a pre-treatment to diminish carbon concentration, followed by aerobic treatment. The AAS was demonstrated to remove more than 85% of ARGs in the inflow compared to that removed by utilizing either aerobic or anaerobic units (Christgen et al. 2015). The sequential operation of aerobic and anaerobic biological treatment in WWTP is competent in the deduction of antibiotics, ARB, and ARGs in contrast to aerobic reactors alone. Moreover, membrane-based technologies like MBR can be used in consolidation with biological treatment to enhance the ARG elimination capability of WWTPs.

Chemical and physical removal method

Constructed wetlands are an attractive wastewater treatment approach for removing antibiotic residues and ARB/ARGs from raw domestic wastewater by the substrate (zeolite, shucks, volcanic rock) adsorption and biodegradation. They are small semi-aquatic ecosystems and are characterized by various physiochemical reactions and multiplication of a

considerable number of the microbial population. The constructed wetlands can be designed with different flow configurations, flow types (surface flow, horizontal subsurface flow, and vertical subsurface flow), and/or plant species (Chen et al. 2016). The efficiency of removal of antibiotic residues attained 75.8 to 98.6%, and the removal rate of total ARGs varied between 63.9 and 84.0% using the constructed wetland. The use of plants is found to be beneficial. Especially for antibiotics removal, the subsurface flow configuration had more potential than the surface flow constructed wetlands (Chen et al. 2016). Fang et al. (2017) studied the removal efficiency of ARGs by the integrated surface flow constructed wetlands in two seasons. The removal rates for 14 ARGs were 77.8% and 59.5% in the winter and summer seasons, respectively.

Disinfection processes have been demonstrated to kill a significant percentage of the microbial population present in wastewater. Yuan et al. (2015) investigated the efficiency of chlorination treatment for removing nine ARBs and eight ARGs (*ere(A)*, *ere(B)*, *erm(A)*, *erm(B)*, *tet(A)*, *tet(B)*, *tet(M)*, and *tet(O)*) in water. The dose of chlorine ranging from 15 to 300 mg-min/L was applied, and the ARG removal percentage examined by quantitative real-time PCR varied from 20 to 60%, respectively. Except sulfadiazine- and erythromycin-resistant bacteria, all other bacterial populations were inactivated completely by just 15 mg-min/L. Sulfadiazine- and erythromycin-resistant bacteria were inactivated at greater than 60 mg of chlorine dose (min/L) (Yuan et al. 2015). A recent study compared the efficiency of three disinfection methods, including chlorination, single UV irradiation, and sequential UV/chlorination, to eliminate ARGs from municipal WWTP (Sharma et al. 2016). The removal of four ARGs (*sul(I)*, *tet(X)*, *tet(G)*, *intI(1)*), and 16S rRNA gene were explored. The application of 30 mg-min/L of Cl_2 showed maximum inactivation in the range from 1.30-log to 1.49-log (Sharma et al. 2016). The sequential UV/chlorination was more effective in ARG elimination, compared to only UV irradiation disinfection.

Recent studies have explored nanomaterials for ARB removal due to their antibacterial properties. The physicochemical properties of nanoparticles and the kind of bacterial targets are two fundamental elements that affect the antibacterial activities of nanoparticles. Aruguete et al. (2013) formulated two types of nanoparticles: a mixture of liposomes, dendrimers, and an antibiotic placed inside a nanoparticle polymer and inorganic nanoparticles with antibiotic molecules affixed to its surfaces. The nanoparticles were made of silver and were coated with a water-soluble polymer called polyvinylpyrrolidone. There are two possible mechanisms for the action of nanoparticle–antibiotic combination on the elimination of ARB/ARGs; the first one is a combination of functionalized nanomaterial and antibiotics entering the cytoplasm of ARB and releasing substantial quantities of ions that cause

cell death and the second includes the synergistic effects of antibiotics and nanomaterials in combating ARGs (Aruguete et al. 2013). In this experiment, the *Pseudomonas aeruginosa* bacteria, which was found resistant to multiple drugs, were proved to be eliminated by this nanomaterial combination (Aruguete et al. 2013).

Recently, Li et al. (2017) investigated the technique of coagulation for its usefulness to eliminate ARGs from wastewater (Li et al. 2017). The coagulation method is known to remove colloidal particles from water actively. The coagulant neutralizes colloidal particles as they carry opposite charges. The coagulants have a positive charge, whereas colloidal particles are mostly negatively charged. The inorganic coagulant FeCl_3 and inorganic polymer coagulant poly ferric chloride (PFC) were inspected for the exclusion of *sul* genes, *tet* genes, and integrase genes. The decrease in dissolved $\text{NH}_3\text{-N}$ and dissolved organic carbon and ARGs were determined to be significantly positively correlated. ARG removal proficiency ranged from 0.5-log to 3.1-log reductions (Li et al. 2017).

The addition of biochar amendments in the soil has been found to exhibit a major modification in the microbial population and inactivate ARGs. Biochar is active charcoal derived from the pyrolysis of biomass rich in carbon. It has high porosity that explains its large specific surface area and is rich in mineral elements (Ye et al. 2016). Thus, the critical mechanism for treatment by biochar is the sorption of a large number of contaminants into the sites present in biochar. The sorption depends on the properties of contaminants, which will determine the dominance of micropore-filling or macropore-filling sorption. Apart from filling, an electrostatic repulsion mechanism can be credited for complex characteristics of biochar. Ye et al. (2016) studied the fate of lettuce plant cultivated on soil contaminated by sulfonamides. The decrease in the concentration of sulfonamides in soil (from 2.9 to 2.2 mg kg^{-1}) and the maximum growth indices (132 to 152 g) were revealed following the biochar amendment in soil. Moreover, compared to vegetables grown in non-amended soil, the reduction of sulfonamide-resistant bacterial endophytes in roots and an increase in the dissipation of sulfonamides differed by one- to twofold were observed in lettuce grown in biochar added to the soil (Ye et al. 2016).

Global policy execution

Large quantities of antibiotics are used in aquaculture, animal husbandry, and agriculture, which exceeds antibiotic use for human health purposes. In Indian context, the antibiotics have been utilized in a behemoth quantity, mainly due to the lack of skilled medical practitioners in primarily rural areas, and no program for regular monitoring for antibiotic use

(Laxminarayan et al. 2013). India and other low- to middle-income countries have no legal regulations for the use of antibiotics as growth promoters, despite its ban in the EU in 2006 (Gonzalez Ronquillo and Angeles Hernandez 2017). The countries in the European Union strictly follow the norms set by European pharmacopeia for the addition of the amount of antibiotics in a pharmaceutical composition. On the other hand, there are no such legal norms in low- and middle-income countries. High amounts of antibiotics are released in water bodies, surface water, and groundwater that has caused the emergence of antibiotic-resistant superbugs. The antibiotics should not be purchased without a prescription. No country in the world has legal norms that put a limit on antibiotic levels in WWTP plant effluent (Grenni et al. 2018). Policies are required to improve the use of antibiotic use as prophylactic in aquaculture.

Bioactive properties of antibiotics further exaggerate the dispute over water security. In high-income countries EU and the USA, the failure of antibiotic effectiveness has reported 25,000 and 23,000 deaths, respectively, while in India, antibiotic resistance has caused 58,000 deaths, according to CDC 2015 reports. The CDC 2015 report suggests a reduction in antibiotic overuse by creating awareness among the general public and policymakers. The CDDEP (2015) has reported a decline in antibiotic prescription rate per capita outpatient from 5 to 0.9% from 1999 to 2012 in the USA (CDDEP 2015).

The “Global Action Plan on Antimicrobial Resistance” outlined by WHO in 2015 provides a framework for the development of national action plans for countering antibiotic resistance crisis (Founou et al. 2016). The aim is to ensure the continued effectiveness of antibiotic drugs for the treatment of infectious diseases. The objectives of “Global Action Plan” are to create public awareness through educating people, bolster the knowledge base through vigilance and research, decrease the outbreak of infection, responsible use of quality-assured antibiotic drugs, and ensure appropriate allocation of the financial budget for the development of replacement medicine to counter infections due to antibiotic resistance (van Rijn et al. 2019). The composition of antibiotics in pharmaceutical products is according to “European pharmacopeia” and are stringently followed by companies in the western world. However, India has no norms that are referred to as antibiotic drug preparation. Furthermore, the multidrug resistance superbugs emergence is exaggerated by the absence of strictness, expertise, and lack of regular monitoring of antibiotic use. The USA has made a “National action plan” based on a framework suggested by the “Global Action Plan” to check the spread of antibiotic-resistant bacteria through supervision (House 2015).

The “European Surveillance of Antimicrobial Consumption (ESVAC)” is a program formed by the “European Medicines Agency (EMA)” to assimilate data regarding animal usage of

antibiotics and variation in demographics of animals that gives information which is precise and reliable. Thus, the ESVAC program entails responsibility for the collection, examination, and report of drug sales data from member states under the EU (EVSAC 2016). Due to that proper restrictions over-controlled the use of antibiotics, the consumption of it has consistently remained constant in high-income countries (CDDEP 2015). The Southeast Asian countries, during the Jaipur declaration, have undertaken stewardship for resistance to antibiotics (WHO, 2015). Policies need to be updated for improving fish hygiene and controlling an outbreak of epizootic diseases, and alleviating prophylactic use of antimicrobials. These new policies should also add limitations to the use of those antibiotics in aquaculture that is used for human medications (Aarestrup et al. 2008). It is predicted that with unchanged continuation policies in BRICS countries, antibiotic consumption might increase twofold by 2030 (Van Boeckel et al. 2015).

Conclusion and future perspective

The antibiotic consumption in BRICS countries has significantly increased in recent years, and to date, around 250 antibiotic drugs are registered for use. The prophylactic use of antibiotics has increased the potential of resistance genes for transfer. The resistance genes in human clinical pathogens are found to be flanked by MGEs, suggesting their transfer through HGT events. It is perceived that the WWTPs are inefficient in the complete removal of antibiotic residues and antibiotic determinants from the water matrices; hence, further exploration on capacities and capabilities of WWTPs is the call of present conditions worldwide. The effect of stress conditions in aerobic/anaerobic digesters on HGT events and fate of ARGs should be studied. The acidic pH and high temperature have been shown to increase ARB/ARG removal efficiency in sludge treatment, but it has not been extrapolated to substantial scale wastewater treatment.

Various environmental facets are responsible for the spread of resistance. The stimuli that cause the bacteria to be competent by natural transformation are understood poorly. The ecotoxicity of antibiotics is well studied in non-target aquatic organisms; however, no toxicity studies in plants or sediment and soil-inhabiting organisms were performed. The combined toxic effects of antibiotic mixtures should be studied further to assess the actual hazard. The study also points out the scarcity of data regarding the toxic effects of the non-therapeutic (low-concentration) antibiotics towards non-target organisms, and the lifetime exposure assays with aquatic organisms. The additive effect of less toxic antibiotics over multiple generations should also be considered. Moreover, the toxic effects of multiple antibiotics and their mixed effect with other contaminants should be further studied. Bioremediation for the elimination of ARGs should consider the correlation between degradation

and resistance potential of applied microorganisms so that its use will not promote the undesirable dissemination of resistance in nature.

In India, the problem of multidrug resistance is thought to increase dramatically in the coming years if no restriction on misuse and overuse of antibiotics is placed. Lack of skilled medical workers in rural areas, absence of regular monitoring, no legal norms for the amount of antibiotic pharmaceutical formulations, and no regulations to set limits for antibiotic concentration in WWTP effluent are also one of the major concerns. India needs policies in place to control the manufacturing process of antibiotics, quality assessment, availability, and use of antibiotics.

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AM helped in the drafting of the manuscript.

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References

- Aarestrup FM, Wegener HC, Collignon P (2008) Resistance in bacteria of the food chain: epidemiology and control strategies. *Expert Rev Anti Infect Ther* 6:733–750. <https://doi.org/10.1586/14787210.6.5.733>
- Abouzeed YM, Baucheron S, Cloeckaert A (2008) ramR mutations involved in efflux-mediated multidrug resistance in *Salmonella enterica* serovar Typhimurium. *Antimicrob Agents Chemother* 52:2428–2434. <https://doi.org/10.1128/AAC.00084-08>
- Aedo S, Ivanova L, Tomova A, Cabello FC (2014) Plasmid-related quinolone resistance determinants in epidemic vibrio parahaemolyticus, uropathogenic *Escherichia coli*, and marine bacteria from an aquaculture area in Chile. *Microb Ecol* 68:324–328. <https://doi.org/10.1007/s00248-014-0409-2>
- Akiba M, Senba H, Otagiri H, Prabhasankar VP, Taniyasu S, Yamashita N, Lee KI, Yamamoto T, Tsutsui T, Ian Joshua D, Balakrishna K, Bairy I, Iwata T, Kusumoto M, Kannan K, Guruge KS (2015) Impact of wastewater from different sources on the prevalence of antimicrobial-resistant *Escherichia coli* in sewage treatment plants in South India. *Ecotoxicol Environ Saf* 115:203–208. <https://doi.org/10.1016/j.ecoenv.2015.02.018>
- Akici A, Aydin V, Kiroglu A (2018) Assessment of the association between drug disposal practices and drug use and storage behaviors. *Saudi Pharm J* 26:7–13. <https://doi.org/10.1016/j.jsps.2017.11.006>
- Allen HK, Moe LA, Rodbumrer J, Gaarder A, Handelsman J (2009) Functional metagenomics reveals diverse B-lactamases in a remote Alaskan soil. *ISME J* 3:243–251. <https://doi.org/10.1038/ismej.2008.86>
- Alshanti Adli NY (2014) Toxicity of single and mixtures of antibiotics to Cyanobacteria. *J Environ Anal Toxicol* 05. <https://doi.org/10.4172/2161-0525.1000274>
- Andersson DI, Hughes D (2010) Antibiotic resistance and its cost: Is it possible to reverse resistance? *Nat Rev Microbiol* 8:260–271. <https://doi.org/10.1038/nrmicro2319>
- Ao X, Liu W, Sun W, Cai M, Ye Z, Yang C, Lu Z, Li C (2018) Medium pressure UV-activated peroxymonosulfate for ciprofloxacin degradation: kinetics, mechanism, and genotoxicity. *Chem Eng J* 345:87–97. <https://doi.org/10.1016/j.cej.2018.03.133>
- Aruguete DM, Kim B, Hochella MF, Ma Y, Cheng Y, Hoegh A, Liu J, Pruden A (2013) Antimicrobial nanotechnology: its potential for the effective management of microbial drug resistance and implications for research needs in microbial nanotoxicology. *Environ Sci Process Impacts* 15:93–102. <https://doi.org/10.1039/c2em30692a>
- Arya SC, Agarwal N (2011) International travel with acquisition of multi-drug resistant Gram negative bacteria containing the New Delhi metallo-beta-lactamase gene, bla NDM-1. *Travel Med. Infect. Dis.* 9:47–48
- Asche F, Hansen H, Tveteras R, Tveteras S (2010) The Salmon Disease Crisis in Chile. *J Chem Inf Model* 53:1689–1699. <https://doi.org/10.1017/CBO9781107415324.004>
- aus der Beek T, Weber FA, Bergmann A et al (2016) Pharmaceuticals in the environment-global occurrences and perspectives. *Environ Toxicol Chem* 35:823–835. <https://doi.org/10.1002/etc.3339>
- Aydin S, Aydin ME, Ulvi A, Kilic H (2019) Antibiotics in hospital effluents: occurrence, contribution to urban wastewater, removal in a wastewater treatment plant, and environmental risk assessment. *Environ Sci Pollut Res* 26:544–558. <https://doi.org/10.1007/s11356-018-3563-0>
- Bacanli M, Başaran N (2019) Importance of antibiotic residues in animal food. *Food Chem. Toxicol.* 125:462–466
- Backhaus T, Scholze M, Grimme LH (2000) The single substance and mixture toxicity of quinolones to the bioluminescent bacterium *Vibrio fischeri*. *Aquat Toxicol* 49:49–61. [https://doi.org/10.1016/S0166-445X\(99\)00069-7](https://doi.org/10.1016/S0166-445X(99)00069-7)
- Bailey AM, Ivens A, Kingsley R, Cottell JL, Wain J, Piddock LJV (2010) RamA, a member of the AraC/XylS family, influences both virulence and efflux in *Salmonella enterica* serovar typhimurium. *J Bacteriol* 192:1607–1616. <https://doi.org/10.1128/JB.01517-09>
- Balcázar JL, Subirats J, Borrego CM (2015) The role of biofilms as environmental reservoirs of antibiotic resistance. *Front Microbiol* 6:1–9. <https://doi.org/10.3389/fmicb.2015.01216>
- Baquero F, Coque TM, Martínez JL, Aracil-Gisbert S, Lanza VF (2019) Gene transmission in the one health microbiosphere and the channels of antimicrobial resistance. *Front Microbiol* 10:1–14. <https://doi.org/10.3389/fmicb.2019.02892>
- Barancheshme F, Munir M (2018) Strategies to combat antibiotic resistance in the wastewater treatment plants. *Front Microbiol* 8. <https://doi.org/10.3389/fmicb.2017.02603>
- Baroud M, Dandache I, Araj GF, Wakim R, Kanj S, Kanafani Z, Khairallah M, Sabra A, Shehab M, Dbaiho G, Matar GM (2013) Underlying mechanisms of carbapenem resistance in extended-spectrum β -lactamase-producing *Klebsiella pneumoniae* and *Escherichia coli* isolates at a tertiary care centre in Lebanon: Role of OXA-48 and NDM-1 carbapenemases. *Int J Antimicrob Agents* 41:75–79. <https://doi.org/10.1016/j.ijantimicag.2012.08.010>

- Bassil RJ, Bashour II, Sleiman FT, Abou-Jawdeh YA (2013) Antibiotic uptake by plants from manure-amended soils. *J Environ Sci Health Part B* 48:570–574. <https://doi.org/10.1080/03601234.2013.774898>
- Baucheron S, Nishino K, Monchaux I, Canepa S, Maurel MC, Coste F, Roussel A, Cloeckaert A, Giraud E (2014) Bile-mediated activation of the *acrAB* and *tolC* multidrug efflux genes occurs mainly through transcriptional derepression of *ramA* in *Salmonella enterica* serovar Typhimurium. *J Antimicrob Chemother* 69:2400–2406. <https://doi.org/10.1093/jac/dku140>
- Beceiro A, Llobet E, Aranda J, Bengoechea JA, Doumith M, Hornsey M, Dhanji H, Chart H, Bou G, Livermore DM, Woodford N (2011) Phosphoethanolamine modification of lipid A in colistin-resistant variants of *Acinetobacter baumannii* mediated by the *pmrAB* two-component regulatory system. *Antimicrob Agents Chemother* 55:3370–3379. <https://doi.org/10.1128/AAC.00079-11>
- Berendonk TU, Manaia CM, Merlin C et al (2015) Tackling antibiotic resistance: the environmental framework. *Nat Rev Microbiol* 13:310–317. <https://doi.org/10.1038/nrmicro3439>
- Bhattacharya SS, Yadav JS (2017) Microbial P450 enzymes in bioremediation and drug discovery: emerging potentials and challenges. *Curr Protein Pept Sci* 19. <https://doi.org/10.2174/1389203718666161122105750>
- Bhattacharyya A, Haldar A, Bhattacharyya M, Ghosh A (2019) Anthropogenic influence shapes the distribution of antibiotic resistant bacteria (ARB) in the sediment of Sundarban estuary in India. *Sci Total Environ* 647:1626–1639. <https://doi.org/10.1016/j.scitotenv.2018.08.038>
- Białk-Bielińska A, Stolte S, Arning J, Uebers U, Bösch A, Stepnowski P, Matzke M (2011) Ecotoxicity evaluation of selected sulfonamides. *Chemosphere* 85:928–933. <https://doi.org/10.1016/j.chemosphere.2011.06.058>
- Blair JMA, Webber MA, Baylay AJ, Ogbolu DO, Piddock LJV (2015) Molecular mechanisms of antibiotic resistance. *Nat Rev Microbiol* 13:42–51. <https://doi.org/10.1038/nrmicro3380>
- Bouki C, Venieri D, Diamadopoulos E (2013) Detection and fate of antibiotic resistant bacteria in wastewater treatment plants: a review. *Ecotoxicol. Environ. Saf.* 91:1–9
- Braschi I, Blasioli S, Fellet C, Lorenzini R, Garelli A, Pori M, Giacomini D (2013) Persistence and degradation of new β -lactam antibiotics in the soil and water environment. *Chemosphere* 93:152–159. <https://doi.org/10.1016/j.chemosphere.2013.05.016>
- Brown S, Bantar C, Young HK, Amyes SGB (1998) Limitation of *Acinetobacter baumannii* treatment by plasmid-mediated carbapenemase ARI-2. *Lancet* 351:186–187. [https://doi.org/10.1016/S0140-6736\(05\)78210-6](https://doi.org/10.1016/S0140-6736(05)78210-6)
- Brown KD, Kulis J, Thomson B, Chapman TH, Mawhinney DB (2006) Occurrence of antibiotics in hospital, residential, and dairy effluent, municipal wastewater, and the Rio Grande in New Mexico. *Sci Total Environ* 366:772–783. <https://doi.org/10.1016/j.scitotenv.2005.10.007>
- Cabello FC (2006) Heavy use of prophylactic antibiotics in aquaculture: a growing problem for human and animal health and for the environment. *Environ. Microbiol.* 8:1137–1144
- Cabello FC, Godfrey HP, Buschmann AH, Dölz HJ (2016) Aquaculture as yet another environmental gateway to the development and globalisation of antimicrobial resistance. *Lancet Infect Dis* 16:e127–e133. [https://doi.org/10.1016/S1473-3099\(16\)00100-6](https://doi.org/10.1016/S1473-3099(16)00100-6)
- Cantas L, Shah SQA, Cavaco LM et al (2013) A brief multi-disciplinary review on antimicrobial resistance in medicine and its linkage to the global environmental microbiota. *Front Microbiol* 4:1–14. <https://doi.org/10.3389/fmicb.2013.00096>
- Carlson JC, Mabury SA (2006) Dissipation kinetics and mobility of chlortetracycline, tylosin, and monensin in an agricultural soil in Northumberland County, Ontario, Canada. *Environ Toxicol Chem* 25:1–10. <https://doi.org/10.1897/04-657R.1>
- Carvalho IT, Santos L (2016) Antibiotics in the aquatic environments: a review of the European scenario. *Environ Int* 94:736–757. <https://doi.org/10.1016/j.envint.2016.06.025>
- Casewell M, Friis C, Marco E et al (2003) The European ban on growth-promoting antibiotics and emerging consequences for human and animal health. *J Antimicrob Chemother* 52:159–161. <https://doi.org/10.1093/jac/dkg313>
- CDDEP (2015) The state of the world's antibiotics 2015. 6:68–69. doi: <https://doi.org/10.3126/jcmc.v6i4.16721>
- Chakraborty J, Sapkale V, Shah M, Rajput V, Mehete G, Agawane S, Kamble S, Dharm M (2020) Metagenome sequencing to unveil microbial community composition and prevalence of antibiotic and metal resistance genes in hypersaline and hyperalkaline Lonar Lake, India. *Ecol Indic* 110:105827. <https://doi.org/10.1016/j.ecolind.2019.105827>
- Chen J, Ying GG, Wei XD, Liu YS, Liu SS, Hu LX, He LY, Chen ZF, Chen FR, Yang YQ (2016) Removal of antibiotics and antibiotic resistance genes from domestic sewage by constructed wetlands: effect of flow configuration and plant species. *Sci Total Environ* 571:974–982. <https://doi.org/10.1016/j.scitotenv.2016.07.085>
- Chitanand MP, Kadam TA, Gyananath G, Totewad ND, Balhal DK (2010) Multiple antibiotic resistance indexing of coliforms to identify high risk contamination sites in aquatic environment. *Indian J Microbiol* 50:216–220. <https://doi.org/10.1007/s12088-010-0042-9>
- Chitnis V, Chitnis S, Vaidya K, Ravikant S, Patil S, Chitnis DS (2004) Bacterial population changes in hospital effluent treatment plant in central India. *Water Res* 38:441–447. <https://doi.org/10.1016/j.watres.2003.09.038>
- Chowdhury F, Langenkämper G, Grote M (2016) Studies on uptake and distribution of antibiotics in red cabbage. *J Verbraucherschutz Leb* 11:61–69. <https://doi.org/10.1007/s00003-015-1008-y>
- Christgen B, Yang Y, Ahammad SZ, Li B, Rodriguez DC, Zhang T, Graham DW (2015) Metagenomics shows that low-energy anaerobic-aerobic treatment reactors reduce antibiotic resistance gene levels from domestic wastewater. *Environ Sci Technol* 49:2577–2584. <https://doi.org/10.1021/es505521w>
- Chu HY, Sprouffske K, Wagner A (2018) Assessing the benefits of horizontal gene transfer by laboratory evolution and genome sequencing. *BMC Evol Biol* 18:1–21. <https://doi.org/10.1186/s12862-018-1164-7>
- Cong Q, Yuan X, Qu J (2013) A review on the removal of antibiotics by carbon nanotubes. *Water Sci Technol* 68:1679–1687. <https://doi.org/10.2166/wst.2013.420>
- Costanzo SD, Murby J, Bates J (2005) Ecosystem response to antibiotics entering the aquatic environment. In: *Marine Pollution Bulletin*. Pergamon, pp 218–223. <https://doi.org/10.1128/9781555815479.ch12>
- Cruz-Loya M, Kang TM, Lozano NA, Watanabe R, Tekin E, Damoiseaux R, Savage VM, Yeh PJ (2019) Stressor interaction networks suggest antibiotic resistance co-opted from stress responses to temperature. *ISME J* 13:12–23. <https://doi.org/10.1038/s41396-018-0241-7>
- Cummings DE, Archer KF, Arriola DJ, Baker PA, Faucett KG, Laroya JB, Pfeil KL, Ryan CR, Ryan KRU, Zuill DE (2011) Broad dissemination of plasmid-mediated quinolone resistance genes in sediments of two urban coastal wetlands. *Environ Sci Technol* 45:447–454. <https://doi.org/10.1021/es1029206>
- Cytryn E (2013) The soil resistome: The anthropogenic, the native, and the unknown. *Soil Biol Biochem* 63:18–23. <https://doi.org/10.1016/j.soilbio.2013.03.017>
- Dangi AK, Sharma B, Hill RT, Shukla P (2019) Bioremediation through microbes: systems biology and metabolic engineering approach. *Crit Rev Biotechnol* 39:79–98. <https://doi.org/10.1080/07388551.2018.1500997>

- Dantas G, Sommer MOA, Oluwasegun RD, Church GM (2008) Bacteria subsisting on antibiotics. *Science* 320:100–103. <https://doi.org/10.1126/science.1155157>
- Das BK, Behera BK, Chakraborty HJ, Paria P, Gangopadhyay A, Rout AK, Nayak KK, Parida PK, Rai A (2020) Metagenomic study focusing on antibiotic resistance genes from the sediments of River Yamuna. *Gene* 758:144951. <https://doi.org/10.1016/j.gene.2020.144951>
- Dcosta VM, King CE, Kalan L et al (2011) Antibiotic resistance is ancient. *Nature* 477:457–461
- Devarajan N, Köhler T, Sivalingam P, van Delden C, Mulaji CK, Mpiana PT, Ibelings BW, Poté J (2017) Antibiotic resistant *Pseudomonas* spp. in the aquatic environment: a prevalence study under tropical and temperate climate conditions. *Water Res* 115:256–265. <https://doi.org/10.1016/j.watres.2017.02.058>
- Di Cesare A, Fontaneto D, Doppelbauer J, Corno G (2016) Fitness and recovery of bacterial communities and antibiotic resistance genes in urban wastewaters exposed to classical disinfection treatments. *Environ Sci Technol* 50:10153–10161. <https://doi.org/10.1021/acs.est.6b02268>
- Dibner JJ, Richards JD (2005) Antibiotic growth promoters in agriculture: history and mode of action. In: Poultry Science. Poultry Science Association, pp 634–643. <https://doi.org/10.1093/ps/84.4.634>
- Diwan V, Tamhankar AJ, Khandal RK et al (2010) Antibiotics and antibiotic-resistant bacteria in waters associated with a hospital in Ujjain, India. *BMC Public Health* 10:414
- Dolejska M, Villa L, Poirel L, Nordmann P, Carattoli A (2013) Complete sequencing of an IncHI1 plasmid encoding the carbapenemase NDM-1, the ArmA 16s RNA methylase and a resistance-modulation-cell division/multidrug efflux pump. *J Antimicrob Chemother* 68:34–39. <https://doi.org/10.1093/jac/dks357>
- Donato JJ, Moe LA, Converse BJ, Smart KD, Berklein FC, McManus PS, Handelsman J (2010) Metagenomic analysis of apple orchard soil reveals antibiotic resistance genes encoding predicted bifunctional proteins. *Appl Environ Microbiol* 76:4396–4401. <https://doi.org/10.1128/AEM.01763-09>
- Dowson CG, Coffey TJ, Kell C, Whitley RA (1993) Evolution of penicillin resistance in *Streptococcus pneumoniae*; the role of *Streptococcus mitis* in the formation of a low affinity PBP2B in *S. pneumoniae*. *Mol Microbiol* 9:635–643. <https://doi.org/10.1111/j.1365-2958.1993.tb01723.x>
- Du J, Geng J, Ren H et al (2015) Variation of antibiotic resistance genes in municipal wastewater treatment plant with A2O-MBR system. *Environ Sci Pollut Res* 22:3715–3726. <https://doi.org/10.1007/s11356-014-3552-x>
- EMA (2006) Guideline on the environmental risk assessment of medicinal products for human use. Doc ref. EMA/CHMP/SWP/4447/00. European Medicines Agency. Available online at: http://www.ema.europa.eu/docs/en_GB/document_library/Scientific_guideline/2009/10/WC500003978.pdf
- European Union (2006) Ban on antibiotics as growth promoters in animal feed enters into effect. Regulation 1. https://ec.europa.eu/commission/presscorner/detail/en/IP_05_1687. Accessed 10 Mar 2021
- EVSAC (2016) Sales of veterinary antimicrobial agents in 30 European countries in 2016– Trends from 2010 to 2016– Eight EVSAC report. Eur Med Agency 176. doi:<https://doi.org/10.1016/j.phpro.2011.05.019>
- Fang H, Zhang Q, Nie X, Chen B, Xiao Y, Zhou Q, Liao W, Liang X (2017) Occurrence and elimination of antibiotic resistance genes in a long-term operation integrated surface flow constructed wetland. *Chemosphere* 173:99–106. <https://doi.org/10.1016/j.chemosphere.2017.01.027>
- FDA (2016) Summary report on antimicrobials sold or distributed for use in food-producing animals 6:1689–1699. <https://doi.org/10.1017/CBO9781107415324.004>
- Fick J, Söderström H, Lindberg RH et al (2009) Contamination of surface, ground, and drinking water from pharmaceutical production. *Environ Toxicol Chem* 28:2522. <https://doi.org/10.1897/09-073.1>
- Figueroa-Diva RA, Vasudevan D, MacKay AA (2010) Trends in soil sorption coefficients within common antimicrobial families. *Chemosphere* 79:786–793. <https://doi.org/10.1016/j.chemosphere.2010.03.017>
- Flandroy L, Poutahidis T, Berg G, Clarke G, Dao MC, Decaestecker E, Furman E, Haahtela T, Massart S, Plovier H, Sanz Y, Rook G (2018) The impact of human activities and lifestyles on the interlinked microbiota and health of humans and of ecosystems. *Sci. Total Environ.* 627:1018–1038
- Fleming A (2001) On the antibacterial action of cultures of a penicillium, with special reference to their use in the isolation of *B. influenzae*. 1929. *Bull World Health Organ* 79:780–790. <https://doi.org/10.1093/clinids/2.1.129>
- Forsberg KJ, Reyes A, Wang B, Selleck EM, Sommer MOA, Dantas G (2012) The shared antibiotic resistome of soil bacteria and human pathogens. *Science* 337:1107–1111. <https://doi.org/10.1126/science.1220761>
- Förster M, Laabs V, Lamshöft M et al (2009) Sequestration of manure-applied sulfadiazine residues in soils. *Environ Sci Technol* 43:1824–1830. <https://doi.org/10.1021/es8026538>
- Fortt ZA, Cabello CF, Buschmann RA (2007) Residuos de tetraciclina y quinolonas en peces silvestres en una zona costera donde se desarrolla la acuicultura del salmón en Chile. *Rev Chil Infectologia* 24:14–18. <https://doi.org/10.4067/s0716-10182007000100002>
- Founou LL, Founou RC, Essack SY (2016) Antibiotic resistance in the food chain: a developing country-perspective. *Front Microbiol* 7:1–19. <https://doi.org/10.3389/fmicb.2016.01881>
- Francino MP (2016) Antibiotics and the human gut microbiome: dysbioses and accumulation of resistances. *Front. Microbiol.* 6:1543
- Frazaõ N, Sousa A, Lässig M, Gordo I (2019) Horizontal gene transfer overrides mutation in *Escherichia coli* colonizing the mammalian gut. *Proc Natl Acad Sci U S A* 116:17906–17915. <https://doi.org/10.1073/pnas.1906958116>
- Gao P, Munir M, Xagoraki I (2012) Correlation of tetracycline and sulfonamide antibiotics with corresponding resistance genes and resistant bacteria in a conventional municipal wastewater treatment plant. *Sci Total Environ* 421–422:173–183. <https://doi.org/10.1016/j.scitotenv.2012.01.061>
- García-Fernández A, Dionisi AM, Arena S, Iglesias-Torrens Y, Carattoli A, Luzzi I (2018) Human campylobacteriosis in Italy: emergence of multi-drug resistance to ciprofloxacin, tetracycline, and erythromycin. *Front Microbiol* 9:1–8. <https://doi.org/10.3389/fmicb.2018.01906>
- Gaze WH, Krone SM, Joakim Larsson DG et al (2013) Influence of humans on evolution and mobilization of environmental antibiotic resistome. *Emerg Infect Dis* 19:e120871. <https://doi.org/10.3201/eid1907.120871>
- Gillings MR, Gaze WH, Pruden A, Smalla K, Tiedje JM, Zhu YG (2015) Using the class 1 integron-integrase gene as a proxy for anthropogenic pollution. *ISME J* 9:1269–1279. <https://doi.org/10.1038/ismej.2014.226>
- Göbel A, Thomsen A, McArdell CS et al (2005) Occurrence and sorption behavior of sulfonamides, macrolides, and trimethoprim in activated sludge treatment. *Environ Sci Technol* 39:3981–3989. <https://doi.org/10.1021/es048550a>
- Göbel A, McArdell CS, Joss A et al (2007) Fate of sulfonamides, macrolides, and trimethoprim in different wastewater treatment technologies. *Sci Total Environ* 372:361–371. <https://doi.org/10.1016/j.scitotenv.2006.07.039>

- Golet EM, Xifra I, Siegrist H, Alder AC, Giger W (2003) Environmental exposure assessment of fluoroquinolone antibacterial agents from sewage to soil. *Environ Sci Technol* 37:3243–3249. <https://doi.org/10.1021/es0264448>
- Gonzalez Ronquillo M, Angeles Hernandez JC (2017) Antibiotic and synthetic growth promoters in animal diets: review of impact and analytical methods. *Food Control* 72:255–267. <https://doi.org/10.1016/j.foodcont.2016.03.001>
- González-Pleiter M, Gonzalo S, Rodea-Palomares I, Leganés F, Rosal R, Boltes K, Marco E, Fernández-Piñas F (2013) Toxicity of five antibiotics and their mixtures towards photosynthetic aquatic organisms: implications for environmental risk assessment. *Water Res* 47:2050–2064. <https://doi.org/10.1016/j.watres.2013.01.020>
- Gottig S, Gruber TM, Stecher B et al (2015) In vivo horizontal gene transfer of the Carbapenemase OXA-48 during a nosocomial outbreak. *Clin Infect Dis* 60:1808–1815. <https://doi.org/10.1093/cid/civ191>
- Grenni P, Ancona V, Barra Caracciolo A (2018) Ecological effects of antibiotics on natural ecosystems: a review. *Microchem J* 136:25–39. <https://doi.org/10.1016/j.microc.2017.02.006>
- Gros M, Petrović M, Ginebreda A, Barceló D (2010) Removal of pharmaceuticals during wastewater treatment and environmental risk assessment using hazard indexes. *Environ Int* 36:15–26. <https://doi.org/10.1016/j.envint.2009.09.002>
- Gulkowska A, Sander M, Hollender J, Krauss M (2013) Covalent binding of sulfamethazine to natural and synthetic humic acids: assessing laccase catalysis and covalent bond stability. *Environ Sci Technol* 47:6916–6924. <https://doi.org/10.1021/es3044592>
- Gullberg E, Cao S, Berg OG, Ilbäck C, Sandegren L, Hughes D, Andersson DI (2011) Selection of resistant bacteria at very low antibiotic concentrations. *PLoS Pathog* 7:1–10. <https://doi.org/10.1371/journal.ppat.1002158>
- Hakenbeck R, Balmelle N, Weber B et al (2001) Mosaic genes and mosaic chromosomes: intra- and interspecies genomic variation of streptococcus pneumoniae. *Infect Immun* 69:2477–2486. <https://doi.org/10.1128/IAI.69.4.2477-2486.2001>
- Haller MY, Müller SR, McArdell CS et al (2002) Quantification of veterinary antibiotics (sulfonamides and trimethoprim) in animal manure by liquid chromatography-mass spectrometry. *J Chromatogr A* 952:111–120. [https://doi.org/10.1016/S0021-9673\(02\)00083-3](https://doi.org/10.1016/S0021-9673(02)00083-3)
- Halling-Sørensen B (2000) Algal toxicity of antibacterial agents used in intensive farming. *Chemosphere* 40:731–739. [https://doi.org/10.1016/S0045-6535\(99\)00445-2](https://doi.org/10.1016/S0045-6535(99)00445-2)
- Harrabi M, Varela Della Giustina S, Aloulou F, Rodriguez-Mozaz S, Barceló D, Elleuch B (2018) Analysis of multiclass antibiotic residues in urban wastewater in Tunisia. *Environ Nanotechnol Monit Manag* 10:163–170. <https://doi.org/10.1016/j.enmm.2018.05.006>
- Havelkova B, Beklova M, Kovacova V, Hlavkova D, Pikula J (2016) Ecotoxicity of selected antibiotics for organisms of aquatic and terrestrial ecosystems. *Neuroendocrinol Lett* 37:38–44
- He Y, Nurul S, Schmitt H, Sutton NB, Murk TAJ, Blokland MH, Rijnaarts HHM, Langenhoff AAM (2018) Evaluation of attenuation of pharmaceuticals, toxic potency, and antibiotic resistance genes in constructed wetlands treating wastewater effluents. *Sci Total Environ* 631–632:1572–1581. <https://doi.org/10.1016/j.scitotenv.2018.03.083>
- Henríquez P, Kaiser M, Bohle H, Bustos P, Mancilla M (2016) Comprehensive antibiotic susceptibility profiling of Chilean *Piscirickettsia salmonis* field isolates. *J Fish Dis* 39:441–448. <https://doi.org/10.1111/jfd.12427>
- Heuer OE, Kruse H, Grave K, Collignon P, Karunasagar I, Angulo FJ (2009) Human health consequences of use of antimicrobial agents in aquaculture. *Clin Infect Dis* 49:1248–1253. <https://doi.org/10.1086/605667>
- Hillis DG, Lissemore L, Sibley PK, Solomon KR (2007) Effects of monensin on zooplankton communities in aquatic microcosms. *Environ Sci Technol* 41:6620–6626. <https://doi.org/10.1021/es070799f>
- Hirakawa H, Inazumi Y, Masaki T, Hirata T, Yamaguchi A (2005) Indole induces the expression of multidrug exporter genes in *Escherichia coli*. *Mol Microbiol* 55:1113–1126. <https://doi.org/10.1111/j.1365-2958.2004.04449.x>
- Hong B, Lin Q, Yu S, Chen Y, Chen Y, Chiang P (2018) Urbanization gradient of selected pharmaceuticals in surface water at a watershed scale. *Sci Total Environ* 634:448–458. <https://doi.org/10.1016/j.scitotenv.2018.03.392>
- House W (2015) National action plan for combating antibiotic-resistant bacteria. The White House, Washington, DC
- Hu Y, Yang X, Qin J, Lu N, Cheng G, Wu N, Pan Y, Li J, Zhu L, Wang X, Meng Z, Zhao F, Liu D, Ma J, Qin N, Xiang C, Xiao Y, Li L, Yang H, Wang J, Yang R, Gao GF, Wang J, Zhu B (2013) Metagenome-wide analysis of antibiotic resistance genes in a large cohort of human gut microbiota. *Nat Commun* 4:2151. <https://doi.org/10.1038/ncomms3151>
- Huber MM, Göbel A, Joss A et al (2005) Oxidation of pharmaceuticals during ozonation of municipal wastewater effluents: a pilot study. *Environ Sci Technol* 39:4290–4299. <https://doi.org/10.1021/es048396s>
- Huq A, Whitehouse CA, Grim CJ, Alam M, Colwell RR (2008) Biofilms in water, its role and impact in human disease transmission. *Curr Opin Biotechnol* 19:244–247. <https://doi.org/10.1016/j.copbio.2008.04.005>
- Ikehata K, Jodeiri Naghashkar N, Gamal El-Din M (2006) Degradation of aqueous pharmaceuticals by ozonation and advanced oxidation processes: a review. *Ozone Sci Eng* 28:353–414. <https://doi.org/10.1080/01919510600985937>
- Ingerslev F, Halling-Sørensen B (2000) Biodegradability properties of sulfonamides in activated sludge. *Environ Toxicol Chem* 19:2467–2473. <https://doi.org/10.1002/etc.5620191011>
- Ioele G, De Luca M, Ragno G (2016) Acute toxicity of antibiotics in surface waters by bioluminescence test. *Curr Pharm Anal* 12:220–226. <https://doi.org/10.2174/1573412912666151110204041>
- Isidori M, Lavorgna M, Nardelli A, Pascarella L, Parrella A (2005) Toxic and genotoxic evaluation of six antibiotics on non-target organisms. *Sci Total Environ* 346:87–98. <https://doi.org/10.1016/j.scitotenv.2004.11.017>
- Jackson RW, Johnson LJ, Clarke SR, Arnold DL (2011) Bacterial pathogen evolution: breaking news. *Trends Genet* 27:32–40. <https://doi.org/10.1016/j.tig.2010.10.001>
- Jechalke S, Heuer H, Siemens J, Amelung W, Smalla K (2014) Fate and effects of veterinary antibiotics in soil. *Trends Microbiol* 22:536–545. <https://doi.org/10.1016/j.tim.2014.05.005>
- Jiang JQ, Zhou Z, Sharma VK (2013) Occurrence, transportation, monitoring and treatment of emerging micro-pollutants in waste water - a review from global views. *Microchem. J.* 110:292–300
- Jones OAH, Voulvoulis N, Lester JN (2002) Aquatic environmental assessment of the top 25 English prescription pharmaceuticals. *Water Res* 36:5013–5022. [https://doi.org/10.1016/S0043-1354\(02\)00227-0](https://doi.org/10.1016/S0043-1354(02)00227-0)
- Kalasseri SG, Krishnan R, Vattiringal RK, Paul R, Mathew P, Pillai D (2020) Detection of New Delhi metallo-β-lactamase 1 and cephalosporin resistance genes among carbapenem-resistant Enterobacteriaceae in water bodies adjacent to hospitals in India. *Curr Microbiol.* 77:2886–2895. <https://doi.org/10.1007/s00284-020-02107-y>
- Kang DH, Gupta S, Rosen C, Fritz V, Singh A, Chander Y, Murray H, Rohwer C (2013) Antibiotic uptake by vegetable crops from manure-applied soils. *J Agric Food Chem* 61:9992–10001. <https://doi.org/10.1021/jf404045m>
- Karkman A, Pärnänen K, Larsson DGJ (2019) Fecal pollution can explain antibiotic resistance gene abundances in anthropogenically

- impacted environments. *Nat Commun* 10:1–8. <https://doi.org/10.1038/s41467-018-07992-3>
- Karthikeyan KG, Meyer MT (2006) Occurrence of antibiotics in wastewater treatment facilities in Wisconsin, USA. *Sci Total Environ* 361: 196–207. <https://doi.org/10.1016/j.scitotenv.2005.06.030>
- Kim SC, Carlson K (2006) Occurrence of ionophore antibiotics in water and sediments of a mixed-landscape watershed. *Water Res* 40: 2549–2560. <https://doi.org/10.1016/j.watres.2006.04.036>
- Kim S, Jensen JN, Aga DS, Weber AS (2007) Tetracycline as a selector for resistant bacteria in activated sludge. *Chemosphere* 66:1643–1651. <https://doi.org/10.1016/j.chemosphere.2006.07.066>
- Klein EY, Van Boeckel TP, Martinez EM et al (2018) Global increase and geographic convergence in antibiotic consumption between 2000 and 2015. *Proc Natl Acad Sci U S A* 115:E3463–E3470. <https://doi.org/10.1073/pnas.1717295115>
- Klümper U, Riber L, Dechesne A, Sannazzarro A, Hansen LH, Sørensen SJ, Smets BF (2015) Broad host range plasmids can invade an unexpectedly diverse fraction of a soil bacterial community. *ISME J* 9:934–945. <https://doi.org/10.1038/ismej.2014.191>
- Kojima S, Nikaido H (2013) Permeation rates of penicillins indicate that *Escherichia coli* porins function principally as nonspecific channels. *Proc Natl Acad Sci U S A* 110:2629–2634. <https://doi.org/10.1073/pnas.1310333110>
- Kookana RS, Williams M, Boxall ABA, Larsson DGJ, Gaw S, Choi K, Yamamoto H, Thatikonda S, Zhu YG, Carriquiriborde P (2014) Potential ecological footprints of active pharmaceutical ingredients: an examination of risk factors in low-, middle- and high-income countries. *Philos Trans R Soc B Biol Sci* 369:20130586. <https://doi.org/10.1098/rstb.2013.0586>
- Korada SK, Yarla NS, Putta S, et al (2018) A critical appraisal of different food safety and quality management tools to accomplish food safety. In: *Food Safety and Preservation*. Elsevier, pp 1–12. <https://doi.org/10.1016/B978-0-12-814956-0.00001-9>
- Kotwani A, Holloway K (2011) Trends in antibiotic use among outpatients in New Delhi, India. *BMC Infect Dis* 11. <https://doi.org/10.1186/1471-2334-11-99>
- Kovalakova P, Cizmas L, McDonald TJ et al (2020) Occurrence and toxicity of antibiotics in the aquatic environment: A review. *Chemosphere* 251:126351. <https://doi.org/10.1016/j.chemosphere.2020.126351>
- Kumar K, Gupta CS, Chander Y, Singh AK (2005) Antibiotic use in agriculture and its impact on the terrestrial environment. *Adv Agron* 87:1–54. [https://doi.org/10.1016/S0065-2113\(05\)87001-4](https://doi.org/10.1016/S0065-2113(05)87001-4)
- Kumar M, Jaiswal S, Sodhi KK et al (2019) Antibiotics bioremediation: Perspectives on its ecotoxicity and resistance. *Environ Int* 124:448–461. <https://doi.org/10.1016/j.envint.2018.12.065>
- Kumar RR, Lee JT, Cho JY (2012) Fate, occurrence, and toxicity of veterinary antibiotics in environment. *J Korean Soc Appl Biol Chem* 55:701–709. <https://doi.org/10.1007/s13765-012-2220-4>
- Kumar M, Sulfikar, Chaminda T et al (2020) Prevalence of antibiotic resistance in the tropical rivers of Sri Lanka and India. *Environ Res* 188:109765. <https://doi.org/10.1016/j.envres.2020.109765>
- Kumarasamy KK, Toleman MA, Walsh TR, Bagaria J, Butt F, Balakrishnan R, Chaudhary U, Doumith M, Giske CG, Irfan S, Krishnan P, Kumar AV, Maharjan S, Mushtaq S, Noorie T, Paterson DL, Pearson A, Perry C, Pike R, Rao B, Ray U, Sama JB, Sharma M, Sheridan E, Thirunarayan MA, Turton J, Upadhyay S, Warner M, Welfare W, Livermore DM, Woodford N (2010a) Emergence of a new antibiotic resistance mechanism in India, Pakistan, and the UK: a molecular, biological, and epidemiological study. *Lancet Infect Dis* 10:597–602. [https://doi.org/10.1016/S1473-3099\(10\)70143-2](https://doi.org/10.1016/S1473-3099(10)70143-2)
- Kumarasamy KK, Toleman MA, Walsh TR, Bagaria J, Butt F, Balakrishnan R, Chaudhary U, Doumith M, Giske CG, Irfan S, Krishnan P, Kumar AV, Maharjan S, Mushtaq S, Noorie T, Paterson DL, Pearson A, Perry C, Pike R, Rao B, Ray U, Sama JB, Sharma M, Sheridan E, Thirunarayan MA, Turton J, Upadhyay S, Warner M, Welfare W, Livermore DM, Woodford N (2010b) Emergence of a new antibiotic resistance mechanism in India, Pakistan, and the UK: a molecular, biological, and epidemiological study. *Lancet Infect Dis* 10:597–602. [https://doi.org/10.1016/S1473-3099\(10\)70143-2](https://doi.org/10.1016/S1473-3099(10)70143-2)
- Kümmerer K (2009) Antibiotics in the aquatic environment - a review - Part I. *Chemosphere* 75:417–434. <https://doi.org/10.1016/j.chemosphere.2008.11.086>
- Lamba M, Sreekrishnan TR, Ahammad SZ (2020) Sewage mediated transfer of antibiotic resistance to River Yamuna in Delhi, India. *J Environ Chem Eng* 8:102088. <https://doi.org/10.1016/j.jece.2017.12.041>
- Larsson DGJ (2014) Antibiotics in the environment. *Ups. J. Med. Sci.* 119:108–112
- Larsson DGJ, de Pedro C, Paxeus N (2007) Effluent from drug manufactures contains extremely high levels of pharmaceuticals. *J Hazard Mater* 148:751–755. <https://doi.org/10.1016/j.jhazmat.2007.07.008>
- Laskaris P, Tolba S, Calvo-Bado L, Wellington L (2010) Coevolution of antibiotic production and counter-resistance in soil bacteria. *Environ Microbiol* 12:783–796. <https://doi.org/10.1111/j.1462-2920.2009.02125.x>
- Laxminarayan R, Duse A, Wattal C, Zaidi AKM, Wertheim HFL, Sumpradit N, Vlieghe E, Hara GL, Gould IM, Goossens H, Greko C, So AD, Bigdeli M, Tomson G, Woodhouse W, Ombaka E, Peralta AQ, Qamar FN, Mir F, Kariuki S, Bhutta ZA, Coates A, Bergstrom R, Wright GD, Brown ED, Cars O (2013) Antibiotic resistance—the need for global solutions. *Lancet Infect Dis* 13: 1057–1098. [https://doi.org/10.1016/S1473-3099\(13\)70318-9](https://doi.org/10.1016/S1473-3099(13)70318-9)
- Le Hello S, Hendriksen RS, Doublet B et al (2011) International spread of an epidemic population of *Salmonella enterica* serotype Kentucky ST198 resistant to ciprofloxacin. *J Infect Dis* 204:675–684. <https://doi.org/10.1093/infdis/jir409>
- Leung HW, Minh TB, Murphy MB, Lam JCW, So MK, Martin M, Lam PKS, Richardson BJ (2012) Distribution, fate and risk assessment of antibiotics in sewage treatment plants in Hong Kong, South China. *Environ Int* 42:1–9. <https://doi.org/10.1016/j.envint.2011.03.004>
- Li N, Sheng GP, Lu YZ, Zeng RJ, Yu HQ (2017) Removal of antibiotic resistance genes from wastewater treatment plant effluent by coagulation. *Water Res* 111:204–212. <https://doi.org/10.1016/j.watres.2017.01.010>
- Liu L, Chen X, Skogerboe G, Zhang P, Chen R, He S, Huang DW (2012) The human microbiome: a hot spot of microbial horizontal gene transfer. *Genomics* 100:265–270. <https://doi.org/10.1016/j.ygeno.2012.07.012>
- Liu Y, Wang F, Chen X, Zhang J, Gao B (2015) Cellular responses and biodegradation of amoxicillin in *Microcystis aeruginosa* at different nitrogen levels. *Ecotoxicol Environ Saf* 111:138–145. <https://doi.org/10.1016/j.ecoenv.2014.10.011>
- Löffler D, Ternes TA (2003) Analytical method for the determination of the aminoglycoside gentamicin in hospital wastewater via liquid chromatography-electrospray-tandem mass spectrometry. *J Chromatogr A* 1000:583–588. [https://doi.org/10.1016/S0021-9673\(03\)00059-1](https://doi.org/10.1016/S0021-9673(03)00059-1)
- Long KS, Poehlsgaard J, Kehrenberg C, Schwarz S, Vester B (2006) The Cfr rRNA methyltransferase confers resistance to phenicols, lincosamides, oxazolidinones, pleuromutilins, and streptogramin A antibiotics. *Antimicrob Agents Chemother* 50:2500–2505. <https://doi.org/10.1128/AAC.00131-06>
- MacPherson DW, Gushulak BD, Baine WB et al (2009) Population mobility, globalization, and antimicrobial drug resistance. *Emerg Infect Dis* 15:1727–1732. <https://doi.org/10.3201/eid1511.090419>
- Manaiá CM, Macedo G, Fatta-Kassinos D, Nunes OC (2016) Antibiotic resistance in urban aquatic environments: can it be controlled? *Appl. Microbiol. Biotechnol.* 100:1543–1557

- Marano RBM, Cytryn E (2018) The mobile resistome in wastewater treatment facilities and downstream environments: introduction horizontal gene transfer in prokaryotes. 129–155. <https://doi.org/10.1002/9781119192428.ch8>
- Marshall BM, Levy SB (2011) Food animals and antimicrobials: impacts on human health. *Clin Microbiol Rev* 24:718–733. <https://doi.org/10.1128/CMR.00002-11>
- Marti E, Variatza E, Balcazar JL (2014) The role of aquatic ecosystems as reservoirs of antibiotic resistance. *Trends Microbiol* 22:36–41. <https://doi.org/10.1016/j.tim.2013.11.001>
- Massé DI, Saady NMC, Gilbert Y (2014) Potential of biological processes to eliminate antibiotics in livestock manure: an overview. *Animals* 4:146–163. <https://doi.org/10.3390/ani4020146>
- Meena VD, Dotaniya ML, Saha JK, Patra AK (2011) Microbial and physico-chemical changes in tomato juice subjected to pulsed electric field treatment. *Afr J Agric Res* 6:6348–6353. <https://doi.org/10.5897/A>
- Meibom KL, Blokesch M, Dolganov NA, Wu CY, Schoolnik GK (2005) Microbiology: chitin induces natural competence in *Vibrio cholerae*. *Science* 310:1824–1827. <https://doi.org/10.1126/science.1120096>
- Migliore L, Civitareale C, Brambilla G, Dojmi Di Delupis G (1997) Toxicity of several important agricultural antibiotics to *Artemia*. *Water Res* 31:1801–1806. [https://doi.org/10.1016/S0043-1354\(96\)00412-5](https://doi.org/10.1016/S0043-1354(96)00412-5)
- Milić N, Milanović M, Letić NG, Sekulić MT, Radonić J, Mihajlović I, Miloradov MV (2013) Occurrence of antibiotics as emerging contaminant substances in aquatic environment. *Int J Environ Health Res* 23:296–310. <https://doi.org/10.1080/09603123.2012.733934>
- Minguez L, Pedelucq J, Farcy E et al (2016) Toxicities of 48 pharmaceuticals and their freshwater and marine environmental assessment in northwestern France. *Environ Sci Pollut Res* 23:4992–5001. <https://doi.org/10.1007/s11356-014-3662-5>
- Mittal P, Prasodanan PKV, Dhakan DB et al (2019) Metagenome of a polluted river reveals a reservoir of metabolic and antibiotic resistance genes. *Environ Microbiome* 14:5. <https://doi.org/10.1186/s40793-019-0345-3>
- Moreira NFF, Narciso-da-Rocha C, Polo-López MI, Pastrana-Martínez LM, Faria JL, Manaia CM, Fernández-Ibáñez P, Nunes OC, Silva AMT (2018) Solar treatment (H₂O₂, TiO₂-P25 and GO-TiO₂ photocatalysis, photo-Fenton) of organic micropollutants, human pathogen indicators, antibiotic resistant bacteria and related genes in urban wastewater. *Water Res* 135:195–206. <https://doi.org/10.1016/j.watres.2018.01.064>
- Munir M, Wong K, Xagorarakis I (2011) Release of antibiotic resistant bacteria and genes in the effluent and biosolids of five wastewater utilities in Michigan. *Water Res* 45:681–693. <https://doi.org/10.1016/j.watres.2010.08.033>
- Mutiyar PK, Mittal AK (2014) Risk assessment of antibiotic residues in different water matrices in India: key issues and challenges. *Environ Sci Pollut Res* 21:7723–7736. <https://doi.org/10.1007/s11356-014-2702-5>
- Narciso-Da-Rocha C, Varela AR, Schwartz T et al (2014) *bla*TEM and *vanA* as indicator genes of antibiotic resistance contamination in a hospital-urban wastewater treatment plant system. *J Glob Antimicrob Resist* 2:309–315. <https://doi.org/10.1016/j.jgar.2014.10.001>
- Navarrete P, Mardones P, Opazo R, Espejo R, Romero J (2008) Oxytetracycline treatment reduces bacterial diversity of intestinal microbiota of Atlantic salmon. *J Aquat Anim Health* 20:177–183. <https://doi.org/10.1577/H07-043.1>
- Newton DF, MacFarlane S, MacFarlane GT (2013) Effects of antibiotics on bacterial species composition and metabolic activities in chemostats containing defined populations of human gut microorganisms. *Antimicrob Agents Chemother* 57:2016–2025. <https://doi.org/10.1128/AAC.00079-13>
- Nie X, Wang X, Chen J et al (2008) Response of the freshwater alga *Chlorella vulgaris* to trichloroisocyanuric acid and ciprofloxacin. *Environ Toxicol Chem* 27:168–173. <https://doi.org/10.1897/07-028.1>
- Nie XP, Liu BY, Yu HJ, Liu WQ, Yang YF (2013) Toxic effects of erythromycin, ciprofloxacin and sulfamethoxazole exposure to the antioxidant system in *Pseudokirchneriella subcapitata*. *Environ Pollut* 172:23–32. <https://doi.org/10.1016/j.envpol.2012.08.013>
- Niewold TA (2007) The nonantibiotic anti-inflammatory effect of antimicrobial growth promoters, the real mode of action? A hypothesis. *Poult. Sci.* 86:605–609
- Nimonkar YS, Yadav B, Talreja P, Sharma A, Patil S, Saware SS, Ranade DR, Prakash O (2019) Assessment of the role of wastewater treatment plant in spread of antibiotic resistance and bacterial pathogens. *Indian J Microbiol* 59:261–265. <https://doi.org/10.1007/s12088-019-00793-2>
- Nonaka L, Maruyama F, Suzuki S, Masuda M (2015) Novel macrolide-resistance genes, *mef*(C) and *mph*(G), carried by plasmids from *Vibrio* and *Photobacterium* isolated from sediment and seawater of a coastal aquaculture site. *Lett Appl Microbiol* 61:1–6. <https://doi.org/10.1111/lam.12414>
- Novo A, André S, Viana P, Nunes OC, Manaia CM (2013) Antibiotic resistance, antimicrobial residues and bacterial community composition in urban wastewater. *Water Res* 47:1875–1887. <https://doi.org/10.1016/j.watres.2013.01.010>
- OECD (2010) Test No. 209: activated sludge, respiration inhibition test. OECD Guidel Test Chem Sect 2. *Eff Biot Syst* 209:1–18. <https://doi.org/10.1787/9789264070080-en>
- OECD - Organization for Economic Co-operation and Development (1984) Earthworm, acute toxicity tests. Test Guideline 207. OECD Guidel Test Chem 207:9. <https://doi.org/10.1787/9789264070042-en>
- Olivares J, Bernardini A, Garcia-Leon G, Corona F, B. Sanchez M, Martinez JL (2013) The intrinsic resistome of bacterial pathogens. *Front. Microbiol* 4:103
- Oliveira TS, Al Aukidy M, Verlicchi P (2018) Occurrence of common pollutants and pharmaceuticals in hospital effluents. In: Verlicchi P (eds) *Hospital Wastewaters. The handbook of environmental chemistry*, vol 60. Springer, Cham. https://doi.org/10.1007/978-94-007-9921-9_9
- Ortiz de García SA, Pinto Pinto G, García-Encina PA, Irusta-Mata R (2014) Ecotoxicity and environmental risk assessment of pharmaceuticals and personal care products in aquatic environments and wastewater treatment plants. *Ecotoxicology* 23:1517–1533. <https://doi.org/10.1007/s10646-014-1293-8>
- Otter JA, French GL (2010) Molecular epidemiology of community-associated methicillin-resistant *Staphylococcus aureus* in Europe. *Lancet Infect. Dis.* 10:227–239
- Pagès JM, James CE, Winterhalter M (2008) The porin and the permeating antibiotic: a selective diffusion barrier in Gram-negative bacteria. *Nat. Rev. Microbiol.* 6:893–903
- Palmer AC, Angelino E, Kishony R (2010) Chemical decay of an antibiotic inverts selection for resistance. *Nat Chem Biol* 6:105–107. <https://doi.org/10.1038/nchembio.289>
- Piddock LJ (2006) Multidrug resistance efflux pumps in bacteria. *Clin Microbiol* 19:382–402. <https://doi.org/10.1128/CMR.19.2.382>
- Piña B, Bayona JM, Christou A, Fatta-Kassinos D, Guillón E, Lambropoulou D, Michael C, Polesel F, Sayen S (2020) On the contribution of reclaimed wastewater irrigation to the potential exposure of humans to antibiotics, antibiotic resistant bacteria and antibiotic resistance genes - NEREUS COST Action ES1403 position paper. *J Environ Chem Eng* 8:102131. <https://doi.org/10.1016/j.jece.2018.01.011>
- Poirer L, Liard A, Rodriguez-Martinez JM, Nordmann P (2005) *Vibronaceae* as a possible source of Qnr-like quinolone resistance determinants. *J Antimicrob Chemother* 56:1118–1121. <https://doi.org/10.1093/jac/dki371>

- Poirel L, Cattoir V, Nordmann P (2012) Plasmid-mediated quinolone resistance; interactions between human, animal, and environmental ecologies. *Front Microbiol* 3:1–7. <https://doi.org/10.3389/fmicb.2012.00024>
- Prestinaci F, Pezzotti P, Pantosti A (2015) Antimicrobial resistance: a global multifaceted phenomenon. *Pathog Glob Health* 109:309–318. <https://doi.org/10.1179/204773215Y.0000000030>
- Qin S, Wang Y, Zhang Q, Chen X, Shen Z, Deng F, Wu C, Shen J (2012) Identification of a novel genomic island conferring resistance to multiple aminoglycoside antibiotics in *Campylobacter coli*. *Antimicrob Agents Chemother* 56:5332–5339. <https://doi.org/10.1128/AAC.00809-12>
- Quoc Tuc D, Elodie MG, Pierre L et al (2017) Fate of antibiotics from hospital and domestic sources in a sewage network. *Sci Total Environ* 575:758–766. <https://doi.org/10.1016/j.scitotenv.2016.09.118>
- Radu E, Woegerbauer M, Oismuller M, Kreuzinger N (2017) Impact of antibiotics of anthropogenic origin on bacterial soil communities in agricultural ecosystem. pp 255–264. <https://doi.org/10.21698/simi.2017.0033>
- Read AF, Baigent SJ, Powers C et al (2015) Imperfect vaccination can enhance the transmission of highly virulent pathogens. *PLoS Biol* 13:1–18. <https://doi.org/10.1371/journal.pbio.1002198>
- Reybroeck W, Daeseleire E, De Brabander HF, Herman L (2012) Antimicrobials in beekeeping. *Vet Microbiol* 158:1–11. <https://doi.org/10.1016/j.vetmic.2012.01.012>
- Rhouma M, Beaudry F, Thériault W et al (2015) Gastric stability and oral bioavailability of colistin sulfate in pigs challenged orally with *Escherichia coli* O149: F4 (K88). *Res Vet Sci* 102:173–181. <https://doi.org/10.1016/j.rvsc.2015.08.005>
- Roberts MC (2009) The evolution of antibiotic- and host ecosystems 213–230. <https://doi.org/10.1128/9781555815479.ch12>
- Robinson AA, Belden JB, Lydy MJ (2005) Toxicity of fluoroquinolone antibiotics to aquatic organisms. *Environ Toxicol Chem* 24:423–430. <https://doi.org/10.1897/04-210R.1>
- Roe MT, Pillai SD (2003) Monitoring and identifying antibiotic resistance mechanisms in bacteria. In: *Poultry Science*. Poultry Science Association, pp 622–626. <https://doi.org/10.1093/ps/82.4.622>
- Rogowska J, Zimmermann A, Muszyńska A, Ratajczyk W, Wolska L (2019) Pharmaceutical household waste practices: preliminary findings from a case study in Poland. *Environ Manage* 64:97–106. <https://doi.org/10.1007/s00267-019-01174-7>
- Romanowska J, Reuter N, Trylska J (2013) Comparing aminoglycoside binding sites in bacterial ribosomal RNA and aminoglycoside modifying enzymes. *Proteins Struct Funct Bioinforma* 81:63–80. <https://doi.org/10.1002/prot.24163>
- Rosendahl I, Siemens J, Groeneweg J, Linzbach E, Laabs V, Herrmann C, Vereecken H, Amelung W (2011) Dissipation and sequestration of the veterinary antibiotic sulfadiazine and its metabolites under field conditions. *Environ Sci Technol* 45:5216–5222. <https://doi.org/10.1021/es200326t>
- Saxena P, Hiwrale I, Das S, Shukla V, Tyagi L, Pal S, Dafale N, Dhodapkar R (2021) Profiling of emerging contaminants and antibiotic resistance in sewage treatment plants: an Indian perspective. *J Hazard Mater* 408:124877. <https://doi.org/10.1016/j.jhazmat.2020.124877>
- Schwartz T, Kohnen W, Jansen B, Obst U (2003) Detection of antibiotic-resistant bacteria and their resistance genes in wastewater, surface water, and drinking water biofilms. *FEMS Microbiol Ecol* 43:325–335. [https://doi.org/10.1016/S0168-6496\(02\)00444-0](https://doi.org/10.1016/S0168-6496(02)00444-0)
- Shan J, Yang P, Rahman MM, Shang X, Yan X (2018) Tetracycline and sulfamethazine alter dissimilatory nitrate reduction processes and increase N₂O release in rice fields. *Environ Pollut* 242:788–796. <https://doi.org/10.1016/j.envpol.2018.07.061>
- Sharma VK (2008) Oxidative transformations of environmental pharmaceuticals by Cl₂, ClO₂, O₃, and Fe(VI): kinetics assessment. *Chemosphere* 73:1379–1386. <https://doi.org/10.1016/j.chemosphere.2008.08.033>
- Sharma VK, Johnson N, Cizmas L, McDonald TJ, Kim H (2016) A review of the influence of treatment strategies on antibiotic resistant bacteria and antibiotic resistance genes. *Chemosphere* 150:702–714. <https://doi.org/10.1016/j.chemosphere.2015.12.084>
- Shore AC, Deasy EC, Slickers P, Brennan G, O'Connell B, Monecke S, Ehrlich R, Coleman DC (2011) Detection of staphylococcal cassette chromosome mec type XI carrying highly divergent mecA, mecI, mecR1, blaZ, and ccr genes in human clinical isolates of clonal complex 130 methicillin-resistant *Staphylococcus aureus*. *Antimicrob Agents Chemother* 55:3765–3773. <https://doi.org/10.1128/AAC.00187-11>
- Singh SK, Ekka R, Mishra M, Mohapatra H (2017) Association study of multiple antibiotic resistance and virulence: a strategy to assess the extent of risk posed by bacterial population in aquatic environment. *Environ Monit Assess* 189:320. <https://doi.org/10.1007/s10661-017-6005-4>
- Sitaraman R (2018) Prokaryotic horizontal gene transfer within the human holobiont: Ecological-evolutionary inferences, implications and possibilities 06 Biological Sciences 0604 Genetics. *Microbiome* 6:163
- Smillie CS, Smith MB, Friedman J, Cordero OX, David LA, Alm EJ (2011) Ecology drives a global network of gene exchange connecting the human microbiome. *Nature* 480:241–244. <https://doi.org/10.1038/nature10571>
- Sørum H, L'Abée-Lund TM, Solberg A, Wold A (2003) Integron-containing IncU R plasmids pRAS1 and pAr-32 from the fish pathogen *Aeromonas salmonicida*. *Antimicrob Agents Chemother* 47:1285–1290. <https://doi.org/10.1128/AAC.47.4.1285-1290.2003>
- Spanogiannopoulos P, Waglechner N, Koteva K, Wright GD (2014) A rifamycin inactivating phosphotransferase family shared by environmental and pathogenic bacteria. *Proc Natl Acad Sci U S A* 111:7102–7107. <https://doi.org/10.1073/pnas.1402358111>
- Srinivasan P, Sarmah AK, Manley-Harris M (2013) Co-contaminants and factors affecting the sorption behaviour of two sulfonamides in pasture soils. *Environ Pollut* 180:165–172. <https://doi.org/10.1016/j.envpol.2013.05.022>
- Straub JO (2016) Aquatic environmental risk assessment for human use of the old antibiotic sulfamethoxazole in Europe. *Environ Toxicol Chem* 35:767–779. <https://doi.org/10.1002/etc.2945>
- Tacconelli E, Carrara E, Savoldi A, Harbarth S, Mendelson M, Monnet DL, Pulcini C, Kahlmeter G, Kluytmans J, Carmeli Y, Ouellette M, Outtersen K, Patel J, Cavalieri M, Cox EM, Houchens CR, Grayson ML, Hansen P, Singh N, Theuretzbacher U, Magrini N, Aboderin AO, al-Abri SS, Awang Jilil N, Benzonana N, Bhattacharya S, Brink AJ, Burkert FR, Cars O, Cornaglia G, Dyar OJ, Friedrich AW, Gales AC, Gandra S, Giske CG, Goff DA, Goossens H, Gottlieb T, Guzman Blanco M, Hryniewicz W, Kattula D, Jinks T, Kanj SS, Kerr L, Kieny MP, Kim YS, Kozlov RS, Labarca J, Laxminarayan R, Leder K, Leibovici L, Levy-Hara G, Littman J, Malhotra-Kumar S, Manchanda V, Moja L, Ndoye B, Pan A, Paterson DL, Paul M, Qiu H, Ramon-Pardo P, Rodríguez-Baño J, Sanguinetti M, Sengupta S, Sharland M, Si-Mehand M, Silver LL, Song W, Steinbakk M, Thomsen J, Thwaites GE, van der Meer JWM, van Kinh N, Vega S, Villegas MV, Wechsler-Fördös A, Wertheim HFL, Wesangula E, Woodford N, Yilmaz FO, Zorzet A (2018) Discovery, research, and development of new antibiotics: the WHO priority list of antibiotic-resistant bacteria and tuberculosis. *Lancet Infect Dis* 18:318–327. [https://doi.org/10.1016/S1473-3099\(17\)30753-3](https://doi.org/10.1016/S1473-3099(17)30753-3)
- Tamae C, Liu A, Kim K, Sitz D, Hong J, Becket E, Bui A, Solaimani P, Tran KP, Yang H, Miller JH (2008) Determination of antibiotic hypersensitivity among 4,000 single-gene-knockout mutants of *Escherichia coli*. *J Bacteriol* 190:5981–5988. <https://doi.org/10.1128/JB.01982-07>

- Tasho RP, Cho JY (2016) Veterinary antibiotics in animal waste, its distribution in soil and uptake by plants: a review. *Sci. Total Environ.* 563–564:366–376
- Taylor NGH, Verner-Jeffreys DW, Baker-Austin C (2011) Aquatic systems: maintaining, mixing and mobilising antimicrobial resistance? *Trends Ecol Evol* 26:278–284. <https://doi.org/10.1016/j.tree.2011.03.004>
- Tolls J (2001) Sorption of veterinary pharmaceuticals in soils: a review. *Environ. Sci. Technol.* 35:3397–3406
- Torniaainen K, Tammilehto S, Ulvi V (1996) The effect of pH, buffer type and drug concentration on the photodegradation of ciprofloxacin. *Int J Pharm* 132:53–61. [https://doi.org/10.1016/0378-5173\(95\)04332-2](https://doi.org/10.1016/0378-5173(95)04332-2)
- Tran NH, Reinhard M, Gin KYH (2018) Occurrence and fate of emerging contaminants in municipal wastewater treatment plants from different geographical regions—a review. *Water Res.* 133:182–207
- Unemo M, Golparian D, Nicholas R, Ohnishi M, Gallay A, Sednaoui P (2012) High-level cefixime- and ceftriaxone-resistant *Neisseria gonorrhoeae* in France: novel penA mosaic allele in a successful international clone causes treatment failure. *Antimicrob Agents Chemother* 56:1273–1280. <https://doi.org/10.1128/AAC.05760-11>
- Väitalo P, Kruglova A, Mikola A, Vahala R (2017) Toxicological impacts of antibiotics on aquatic micro-organisms: a mini-review. *Int J Hyg Environ Health* 220:558–569. <https://doi.org/10.1016/j.ijheh.2017.02.003>
- Van Boeckel TP, Gandra S, Ashok A et al (2014) Global antibiotic consumption 2000 to 2010: an analysis of national pharmaceutical sales data. *Lancet Infect Dis* 14:742–750. [https://doi.org/10.1016/S1473-3099\(14\)70780-7](https://doi.org/10.1016/S1473-3099(14)70780-7)
- Van Boeckel TP, Brower C, Gilbert M et al (2015) Global trends in antimicrobial use in food animals. *Proc Natl Acad Sci U S A* 112: 5649–5654. <https://doi.org/10.1073/pnas.1503141112>
- van der Grinten E, Pikkemaat MG, van den Brandhof EJ, Stroomberg GJ, Kraak MHS (2010) Comparing the sensitivity of algal, cyanobacterial and bacterial bioassays to different groups of antibiotics. *Chemosphere* 80:1–6. <https://doi.org/10.1016/j.chemosphere.2010.04.011>
- van Rijn M, Haverkate M, Achterberg P, Timen A (2019) The public uptake of information about antibiotic resistance in the Netherlands. *Public Underst Sci* 28:486–503. <https://doi.org/10.1177/0963662518823701>
- Varela AR, André S, Nunes OC, Manaia CM (2014) Insights into the relationship between antimicrobial residues and bacterial populations in a hospital-urban wastewater treatment plant system. *Water Res* 54:327–336. <https://doi.org/10.1016/j.watres.2014.02.003>
- Vetting MW, Hegde SS, Wang M, Jacoby GA, Hooper DC, Blanchard JS (2011) Structure of QnrB1, a plasmid-mediated fluoroquinolone resistance factor. *J Biol Chem* 286:25265–25273. <https://doi.org/10.1074/jbc.M111.226936>
- Wachino JI, Yoshida H, Yamane K, Suzuki S, Matsui M, Yamagishi T, Tsutsui A, Konda T, Shibayama K, Arakawa Y (2011) SMB-1, a novel subclass B3 metallo- β -lactamase, associated with ISCR1 and a class 1 integron, from a carbapenem-resistant *Serratia marcescens* clinical isolate. *Antimicrob Agents Chemother* 55:5143–5149. <https://doi.org/10.1128/AAC.05045-11>
- Walsh TR, Weeks J, Livermore DM, Toleman MA (2011) Dissemination of NDM-1 positive bacteria in the New Delhi environment and its implications for human health: an environmental point prevalence study. *Lancet Infect Dis* 11:355–362. [https://doi.org/10.1016/S1473-3099\(11\)70059-7](https://doi.org/10.1016/S1473-3099(11)70059-7)
- Watkinson AJ, Murby EJ, Costanzo SD (2007) Removal of antibiotics in conventional and advanced wastewater treatment: Implications for environmental discharge and wastewater recycling. *Water Res* 41: 4164–4176. <https://doi.org/10.1016/j.watres.2007.04.005>
- Wegener HC (2003) Antibiotics in animal feed and their role in resistance development. *Curr Opin Microbiol* 6:439–445. <https://doi.org/10.1016/j.mib.2003.09.009>
- Welch TJ, Evenhuis J, White DG, McDermott PF, Harbottle H, Miller RA, Griffin M, Wise D (2009) IncA/C plasmid-mediated florfenicol resistance in the catfish pathogen *Edwardsiella ictaluri*. *Antimicrob Agents Chemother* 53:845–846. <https://doi.org/10.1128/AAC.01312-08>
- Wellington EMH, Boxall ABA, Cross P, Feil EJ, Gaze WH, Hawkey PM, Johnson-Rollings AS, Jones DL, Lee NM, Otten W, Thomas CM, Williams AP (2013) The role of the natural environment in the emergence of antibiotic resistance in Gram-negative bacteria. *Lancet Infect Dis* 13:155–165. [https://doi.org/10.1016/S1473-3099\(12\)70317-1](https://doi.org/10.1016/S1473-3099(12)70317-1)
- Wen X, Wang Y, Zou Y, Ma B, Wu Y (2018) No evidential correlation between veterinary antibiotic degradation ability and resistance genes in microorganisms during the biodegradation of doxycycline. *Ecotoxicol Environ Saf* 147:759–766. <https://doi.org/10.1016/j.ecoenv.2017.09.025>
- Wollenberger L, Halling-Sørensen B, Kusk KO (2000) Acute and chronic toxicity of veterinary antibiotics to *Daphnia magna*. *Chemosphere* 40:723–730. [https://doi.org/10.1016/S0045-6535\(99\)00443-9](https://doi.org/10.1016/S0045-6535(99)00443-9)
- Wright MS, Baker-Austin C, Lindell AH, Stepanauskas R, Stokes HW, McArthur JV (2008) Influence of industrial contamination on mobile genetic elements: class 1 integron abundance and gene cassette structure in aquatic bacterial communities. *ISME J* 2:417–428. <https://doi.org/10.1038/ismej.2008.8>
- Xiong JQ, Govindwar S, Kurade MB, Paeng KJ, Roh HS, Khan MA, Jeon BH (2019) Toxicity of sulfamethazine and sulfamethoxazole and their removal by a green microalga, *Scenedesmus obliquus*. *Chemosphere* 218:551–558. <https://doi.org/10.1016/j.chemosphere.2018.11.146>
- Yan W, Xiao Y, Yan W et al (2018) The effect of bioelectrochemical systems on antibiotics removal and antibiotic resistance genes: a review. *Chem Eng J* 358:1421–1437. <https://doi.org/10.1016/j.cej.2018.10.128>
- Yang LH, Ying GG, Su HC, Stauber JL, Adams MS, Binet MT (2008) Growth-inhibiting effects of 12 antibacterial agents and their mixtures on the freshwater microalga *Pseudokirchneriella subcapitata*. *Environ Toxicol Chem* 27:1201–1208. <https://doi.org/10.1897/07-471.1>
- Ye M, Sun M, Feng Y, Wan J, Xie S, Tian D, Zhao Y, Wu J, Hu F, Li H, Jiang X (2016) Effect of biochar amendment on the control of soil sulfonamides, antibiotic-resistant bacteria, and gene enrichment in lettuce tissues. *J Hazard Mater* 309:219–227. <https://doi.org/10.1016/j.jhazmat.2015.10.074>
- Yuan Q, Bin, Guo MT, Yang J (2015) Fate of antibiotic resistant bacteria and genes during wastewater chlorination: implication for antibiotic resistance control. *PLoS ONE* 10:1–11. <https://doi.org/10.1371/journal.pone.0119403>
- Yun SH, Jho EH, Jeong S, Choi S, Kal Y, Cha S (2018) Photodegradation of tetracycline and sulfathiazole individually and in mixtures. *Food Chem Toxicol* 116:108–113. <https://doi.org/10.1016/j.fct.2018.03.037>
- Zarei-Baygi A, Harb M, Wang P, Stadler LB, Smith AL (2019) Evaluating antibiotic resistance gene correlations with antibiotic exposure conditions in anaerobic membrane bioreactors. *Environ Sci Technol* 53:3599–3609. <https://doi.org/10.1021/acs.est.9b00798>
- Zarfl C, Klasmeyer J, Matthies M (2009) A conceptual model describing the fate of sulfadiazine and its metabolites observed in manure-amended soils. *Chemosphere* 77:720–726. <https://doi.org/10.1016/j.chemosphere.2009.08.035>
- Zhang T, Li B (2011) Occurrence, transformation, and fate of antibiotics in municipal wastewater treatment plants. *Crit Rev Environ Sci Technol* 41:951–998. <https://doi.org/10.1080/10643380903392692>
- Zhang S, Han B, Gu J, Wang C, Wang P, Ma Y, Cao J, He Z (2015) Fate of antibiotic resistant cultivable heterotrophic bacteria and antibiotic resistance genes in wastewater treatment processes. *Chemosphere* 135:138–145. <https://doi.org/10.1016/j.chemosphere.2015.04.001>

- Zhao WH, Hu ZQ (2013) Epidemiology and genetics of CTX-M extended-spectrum β -lactamases in Gram-negative bacteria. *Crit Rev Microbiol* 39:79–101. <https://doi.org/10.3109/1040841X.2012.691460>
- Zhao JL, Liu YS, Liu WR, Jiang YX, Su HC, Zhang QQ, Chen XW, Yang YY, Chen J, Liu SS, Pan CG, Huang GY, Ying GG (2015) Tissue-specific bioaccumulation of human and veterinary antibiotics in bile, plasma, liver and muscle tissues of wild fish from a highly urbanized region. *Environ Pollut* 198:15–24. <https://doi.org/10.1016/j.envpol.2014.12.026>
- Zhao R, Feng J, Liu J, Fu W, Li X, Li B (2019) Deciphering of microbial community and antibiotic resistance genes in activated sludge reactors under high selective pressure of different antibiotics. *Water Res* 151:388–402. <https://doi.org/10.1016/j.watres.2018.12.034>
- Zuccato E, Castiglioni S, Bagnati R, Melis M, Fanelli R (2010) Source, occurrence and fate of antibiotics in the Italian aquatic environment. *J Hazard Mater* 179:1042–1048. <https://doi.org/10.1016/j.jhazmat.2010.03.110>

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